

Percolation, Random Walks, and Minimal Cutsets on Infinite Graphs

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Abstract

This project studies two fundamental areas of modern probability theory: random walks on graphs and percolation theory. Foundational material is studied from [PO] and [Gri99], covering Bernoulli bond and site percolation, phase transitions, the classical Peierls argument, and the FKG inequality. We then develop the theory of electrical networks and isoperimetric profiles, culminating in a proof that graphs with isoperimetric dimension greater than 2 are uniformly transient.

We provide a thorough exposition of the paper [EST25], which establishes connections between percolation thresholds and transience of random walks on infinite graphs. This paper encapsulates how probabilistic ideas can be used to obtain deterministic structural results for graphs. We conclude this review with a careful discussion about the ‘gap at 1’ phenomenon for the critical parameter on transitive graphs.

We conjecture that the same results of [EST25] are true in the case of site percolation, and attempt to prove them via the same methods. However, we see that this is in fact not possible, and give counterexamples illustrating this. In particular, we show that the prescription of exposed vertex boundaries is too ‘expensive’. We consider the class of line graphs to illustrate the difficulties encountered, and prove intermediate results for graphs of bounded degree. We make an original contribution by deducing a corollary which partially extends results by Panagiotis and Severo [PS23], and partially resolves a conjecture posed by Benjamini and Schramm [BS96].

AI Statement

Generative AI was used as an assistive tool in the preparation of this report, in accordance with the University’s academic integrity guidelines. Its use included: converting cited mathematical material into compilable L^AT_EX; assisting with the writing and commenting of L^AT_EX and Python code; improving the clarity and articulation of explanatory text; and proofreading for spelling, grammatical, and theoretical errors.

All AI-assisted explanations, concepts, and code included in this report have been carefully checked for correctness and fully understood by the author. Where generative AI was used to produce L^AT_EX code, it has not been cited separately, as the underlying mathematical content originates from cited sources. Citations for generative AI have been included only where its contribution was substantial or where content was originally created by the AI, which in practice applies exclusively to Python code.

1 Introduction

Percolation theory originated in 1957 when Broadbent and Hammersley [BH57] asked whether liquid could seep through a porous stone with randomly blocked channels [BR06]. The question emerged from a practical problem Broadbent was working on for the British Coal Utilisation Research Association - he was studying how gas could penetrate through coal mine gas masks. This simple model - randomly deleting edges or vertices from a graph - has become fundamental in probability theory, with applications to network robustness, epidemic spreading, and materials science. The central mathematical question is one of phase transition: at what critical probability p_c does an infinite connected component first appear? In fact, what makes percolation thresholds mathematically interesting is that they encode structural information about graphs, and relate to the interplay between percolation and random walk behavior [EST25].

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2 Preliminaries

Throughout this report, we assume known the standard graph-theoretic structures and concepts such as graphs themselves, subgraphs, and components. We recall the following definitions in particular.

Definition 2.1. Let $G = (V, E)$ be a graph. Any two vertices u, v are said to be connected if there is a path (a sequence of distinct vertices in which consecutive vertices are adjacent) in the graph from u to v . If two vertices u, v are connected, we write $u \leftrightarrow v$, and if there is an infinite path starting at u , we write $u \leftrightarrow \infty$.

Definition 2.2. A graph G is locally-finite if the degree of every vertex is finite.

Consider now $\mathbb{Z}^d$, the d -dimensional lattice. For $x \in \mathbb{Z}^d$ we write x_i for the i th coordinate of x . We can construct a graph known as the d -dimensional cubic lattice by adding edges between all pairs x, y of vertices of $\mathbb{Z}^d$ where $\sum_{i=1}^d |x_i - y_i| = 1$. We denote this lattice by $\mathbb{L}^d$, and we write $\mathbb{Z}^d$ for the set of vertices of $\mathbb{L}^d$, and $\mathbb{E}^d$ for the set of its edges. Hence we simply write $\mathbb{L}^d = (\mathbb{Z}^d, \mathbb{E}^d)$.

2.1 Bernoulli Percolation

Suppose $\mathcal{L}$ is an infinite, connected graph. There are two types of percolation: *bond* and *site*. Initially, we define each type informally and perhaps more intuitively, then visit a more formal definition in the next section along with the idea of *monotone coupling*.

Fix $p \in [0, 1]$. In the *bond percolation* model on $\mathcal{L}$ with parameter p , we generate a random spanning subgraph $G(\mathcal{L}, p) = (V, E)$ of $\mathcal{L}$ by including each edge $e \in E(\mathcal{L})$ with probability p , independently of all other edges. We denote $G(\mathcal{L}, p)$ as the *percolation graph* on $\mathcal{L}$ with parameter p . Given an edge $e \in E(\mathcal{L})$, if $e \in E(G(\mathcal{L}, p))$ we say e is *open*; otherwise we say the edge e is *closed*.

The *site percolation* model is defined similarly, where instead, we generate a random subgraph $G(\mathcal{L}, p) = (V, E)$ of $\mathcal{L}$ by including each vertex $v \in V(\mathcal{L})$ with probability p , independently of all other vertices. In this case, an edge $e \in E(\mathcal{L})$ is included in E if and only if both endpoints are included in V . The notion of openness and closedness of vertices is analogous to that of edges.

It is natural to consider certain properties of our percolation graph $G(\mathcal{L}, p)$. Of particular interest is the accessibility of vertices from one another.

Definition 2.3. The components of the percolation graph $G(\mathcal{L}, p)$ are called (open) clusters. For $x \in V$ we write $\mathcal{C}(x)$ for the cluster containing x , and $C(x)$ for the vertex set of this cluster, which we call the ‘cluster at x ’. Define the percolation probability at x , $\theta_x(p)$, by

$$\theta_x(p) := \mathbb{P}_p[|C(x)| = \infty]. \quad (1)$$

If $\mathcal{L} = \mathbb{L}^d$, translation invariance means that for all vertices x, y , $\theta(x) = \theta(y)$ so without loss of generality we can take x to be the origin. In this case, we simply write $\theta(p)$ instead of $\theta_x(p)$.

2.2 Monotone Coupling

We now describe a more formal definition from [Gri99] of bond percolation on $\mathbb{L}^d$ and the idea of monotone coupling, which establishes the monotonicity of $\theta(p)$ in p , and allows us to introduce the concept of *phase transitions*. Until Section 3.5, we consider only bond percolation.

Consider the following probability space. As the sample space we take $\Omega = \prod_{e \in \mathbb{E}^d} \{0, 1\}$, points of which are represented as $\omega = (\omega(e) : e \in \mathbb{E}^d)$ and called *configurations*; the value $\omega(e) = 0$ corresponds to e being closed, and $\omega(e) = 1$ corresponds to e being open. We take $\mathcal{F}$ to be the σ -algebra of subsets of Ω generated by the finite-dimensional cylinders. A cylinder set is a subset of Ω that depends on the configuration of

only finitely many edges. Such a set can be written as $\{\omega \in \Omega : \omega(e) = a_e \text{ for all } e \in F\}$, where $F \subset \mathbb{E}^d$ is a finite set and $a_e \in \{0, 1\}$ for all $e \in F$.

Finally, we take product measure with density p on $(\Omega, \mathcal{F})$; this is the measure $\mathbb{P}_p = \prod_{e \in \mathbb{E}^d} \mu_e$ where μ_e is Bernoulli measure on $\{0, 1\}$, given by

$$\mu_e(\omega(e) = 0) = 1 - p, \quad \mu_e(\omega(e) = 1) = p.$$

The existence of such a probability measure follows from Kolmogorov's Extension Theorem. We write $\mathbb{E}_p$ for the corresponding expectation operator, with particular care taken to avoid confusion with the edge set $\mathbb{E}^d$.

Remark 2.4. There is a natural partial order on the set Ω of configurations, given by $\omega_1 \leq \omega_2$ if and only if $\omega_1(e) \leq \omega_2(e)$ for all $e \in \mathbb{E}^d$. This ordering of configurations gives rise to the notion of *increasing events*, which play a central role later. $\mathcal{A} \in \mathcal{F}$ is such an event if $\omega_1 \in \mathcal{A}$ with $\omega_1 \leq \omega_2$ implies $\omega_2 \in \mathcal{A}$. The most natural examples of such events are connection events like $\{u \leftrightarrow v\}$; adding open edges to a configuration does not affect paths through edges that are already open.

Suppose that $(X(e) : e \in \mathbb{E}^d)$ is a family of independent random variables indexed by the edge set $\mathbb{E}^d$, where each $X(e)$ is uniformly distributed on $[0, 1]$. We may *couple* together all bond percolation processes on $\mathbb{L}^d$ as p ranges over the interval $[0, 1]$ in the following way. Let p satisfy $0 \leq p \leq 1$ and define η_p by

$$\eta_p(e) = \begin{cases} 1 & \text{if } X(e) < p, \\ 0 & \text{if } X(e) \geq p. \end{cases} \quad (2)$$

We may think of η_p as being the random outcome of the bond percolation process on $\mathbb{L}^d$ with edge-probability p . It is clear that $\eta_{p_1} \leq \eta_{p_2}$ whenever $p_1 \leq p_2$, which is to say that we may couple together the two percolation processes with edge-probabilities p_1 and p_2 in such a way that the set of open edges of the first process is a subset of the set of open edges of the second. That is, this coupling is *monotone*. More generally, as p increases from 0 to 1, the configuration η_p runs through typical configurations of percolation processes with all edge-probabilities. This monotone coupling yields the following theorem from [PO].

Theorem 2.5 (Monotonicity of Percolation Probability). *On $\mathbb{L}^d$, $\theta(p)$ is non-decreasing in p . That is, if $p \leq q$ then $\theta(p) \leq \theta(q)$.*

3 Phase Transitions

3.1 The Existence of a Phase Transition at p_c

It is fundamental to percolation theory that, for some infinite connected graph $\mathcal{L}$, there exists a critical value $p_c = p_c(\mathcal{L})$ of p such that

$$\theta_x(p) \begin{cases} = 0 & \text{if } p < p_c, \\ > 0 & \text{if } p > p_c; \end{cases}$$

$p_c(\mathcal{L})$ is called the *critical probability* and is defined formally by $p_c(\mathcal{L}) := \sup\{p : \theta_x(p) = 0\}$, which is well-defined as a direct consequence of the following theorem from [PO].

Theorem 3.1. *Let V be the set of vertices of $\mathcal{L}$. For any $x \in V$, $y \in V$, and $p \in (0, 1)$, if $\theta_x(p) > 0$ then $\theta_y(p) > 0$.*

Given $p \in [0, 1]$, either $\theta_x(p) = 0$ for all $x \in V$, or $\theta_x(p) > 0$ for all $x \in V$. This means that p_c is well-defined, in the sense that p_c (as defined above) does not depend on the choice of x .

Thus, if a non-trivial critical probability exists, the monotonicity guarantees that this is indeed a threshold for phase transition (existence of an infinite component) within the percolation. Now that we have this notion of critical probability, we can state an important theorem on its non-triviality in the case where $\mathcal{L} = \mathbb{L}^d$.

Theorem 3.2. *If $d \geq 2$ then $0 < p_c(\mathbb{L}^d) < 1$.*

We prove this theorem in two parts, by proving each side of the bound in each of the following two subsections respectively. First, we prove that any infinite and connected graph $\mathcal{L}$ with bounded degree satisfies $p_c(\mathcal{L}) > 0$ [PO], and then that $p_c(\mathbb{L}^d) < 1$ for $d \geq 2$ using *Peierls' argument* [Gri99] - a classic technique in percolation theory.

3.2 $p_c > 0$ for Graphs of Bounded Degree

Let $\mathcal{L}$ be an infinite connected graph.

Theorem 3.3. *Let $D \in \mathbb{N}$. Suppose $\deg(v) \leq D$ for all $v \in V$. Then $p_c(\mathcal{L}) \geq 1/(D - 1)$.*

Proof. Pick $x \in V$. For $n \in \mathbb{N}$, let S_n be the set of paths in $\mathcal{L}$ of length n , starting at x . Then we claim that

$$|S_n| \leq D(D - 1)^{n-1}, \quad (3)$$

Indeed, there are at most D choices for the first step of the path, and then for each subsequent step there are at most $D - 1$ choices. Hence, (3) follows immediately from the multiplication principle.

Now, let $p \in [0, 1]$. For $\gamma \in S_n$ let E_γ be the event that γ is open in $G(\mathcal{L}, p)$, i.e. every edge of γ is in $G(\mathcal{L}, p)$. Also, let $A_n := \cup_{\gamma \in S_n} E_\gamma$, the event that there exists an open path of length n starting from x . Then $A_{n+1} \subset A_n$ for each n , and

$$\bigcap_{n=1}^{\infty} A_n = \{|C(x)| = \infty\}. \quad (4)$$

For each $\gamma \in S_n$ we have $\mathbb{P}[E_\gamma] = p^n$, so by the union bound and (3),

$$\mathbb{P}[A_n] \leq \sum_{\gamma \in S_n} \mathbb{P}[E_\gamma] \leq p^n \sum_{\gamma \in S_n} 1 \leq p^n D(D - 1)^{n-1} = (pD)((D - 1)p)^{n-1}$$

so that if $p < 1/(D - 1)$ then by (4) and downward continuity of $\mathbb{P}$ we have

$$\theta_x(p) = \mathbb{P}[\cap_{n=1}^{\infty} A_n] = \lim_{n \rightarrow \infty} \mathbb{P}[A_n] = 0,$$

and hence $p_c(\mathcal{L}) \geq p$. Hence, $p_c(\mathcal{L}) \geq 1/(D - 1)$. □

This result clearly shows the lower bound in Theorem 3.2, as in the case of $\mathbb{L}^d$, we have $D = 2d$. Hence $p_c(\mathbb{L}^d) \geq 1/(2d - 1) > 0$.

3.3 Peierls' Argument

The proof that $p_c(\mathbb{L}^d) < 1$ for $d \geq 2$ uses a classical technique known as the *Peierls argument*, named after mathematician Rudolf Peierls and his 1936 work [Pei36] on the Ising model, a statistical mechanical model where each site takes one of two possible states. Peierls proved that a phase transition exists by showing that large ‘contours’ (closed loops separating regions of different states) are exponentially unlikely.

This argument can easily be adapted to our percolation setting. We complete the proof in two steps, by first proving the theorem with the adapted argument for the case when $d = 2$, before extending to higher dimensions.

Theorem 3.4. *We have that $p_c(\mathbb{L}^2) < 1$.*

Proof. We show that $\theta(p) > 0$ if p is sufficiently close to 1. Here, the notion of *planar duality* is useful, which we define in the following way.

Definition 3.5. *The dual lattice to $\mathbb{L}^2$, denoted $\mathbb{L}^{2*}$, is the graph obtained by shifting the lattice $\mathbb{L}^2$ by $(1/2, 1/2)$. That is, the vertex set and edge set of $\mathbb{L}^{2*}$ are given by*

$$V(\mathbb{L}^{2*}) = \{z + (1/2, 1/2) : z \in \mathbb{Z}^2\} \quad \text{and} \quad E(\mathbb{L}^{2*}) = \{\{u, v\} : u, v \in V(\mathbb{L}^{2*}), |u_1 - v_1| + |u_2 - v_2| = 1\}.$$

For any edge $e \in E(\mathbb{L}^2)$ we denote by e^* the edge in $E(\mathbb{L}^{2*})$ that shares a common midpoint with e . We say that e^* is dual to e .

Then, given a realisation of $G(\mathbb{L}^2, p) = (V, E)$, we can define a random subgraph G^* of $\mathbb{L}^{2*}$ by setting $G^* = (V(\mathbb{L}^{2*}), E^*)$, where

$$E^* := \{e^* \in E(\mathbb{L}^{2*}) : e \notin E\},$$

i.e. an edge is open in G^* if its dual edge is closed in $G(\mathbb{L}^2, p)$. This is known as the dual configuration of $G(\mathbb{L}^2, p)$. Note that G^* is a realisation of $G(\mathbb{L}^2, 1 - p)$ (shifted by $(\frac{1}{2}, \frac{1}{2})$).

Returning to the proof, suppose that the open cluster containing the origin in $\mathbb{L}^2$ is finite. In such a case, we can find an open cycle in the dual configuration that surrounds this open cluster (and in particular surrounds the origin); see Figure 1 for an illustration. The dual edges to the edges in such a cycle form a *cutset*, a definition to be visited in more detail later. The rigorous proof of this statement can be found in [Gri99], but for now we can satisfy ourselves simply by inspecting the figure.

We now count the number of possible such cycles in the dual. Let $\rho(n)$ be the number of cycles of length n in $\mathbb{L}^{2*}$ which contain the origin in their interior. We may estimate $\rho(n)$ as follows.

Any such cycle must pass through some vertex $(k + \frac{1}{2}, \frac{1}{2})$ with $0 \leq k < n$: firstly, since it must encircle the origin, it must cross both the positive and negative x -axis. Secondly, it cannot pass through $(k + \frac{1}{2}, \frac{1}{2})$ for any $k \geq n$. This follows because the cycle length must be at least the distance between the points where it intersects the positive and negative x -axis. Thus, each such cycle is determined by a *self-avoiding walk* (a walk that visits no vertex more than once) of length $n - 1$ starting from some vertex $(k + \frac{1}{2}, \frac{1}{2})$ where $0 \leq k < n$.

Let $\sigma(n)$ be the number of self-avoiding walks of length n in $\mathbb{L}^{2*}$ that begin at any given vertex, which is well-defined by translation invariance. Since there are at most n possible starting points $(k + \frac{1}{2}, \frac{1}{2})$ with $0 \leq k < n$, we have:

$$\rho(n) \leq n\sigma(n - 1). \tag{5}$$

Similarly to the proof of Theorem 3.3, we can bound $\sigma(n)$ by noticing that the first step of a self-avoiding walk has 4 possibilities and any subsequent step has at most 3 possibilities, so by the Multiplication Principle,

$$\sigma(n) \leq 4 \times 3^{n-1} \text{ for } n \geq 1. \tag{6}$$

Note that $\sigma(0) = 1$ trivially so the bound is actually true for $n \geq 0$, which is useful below.

Let γ be a cycle in the dual containing the origin in its interior, and let $M(n)$ be the number of such open cycles of length n . By (5) and (6),

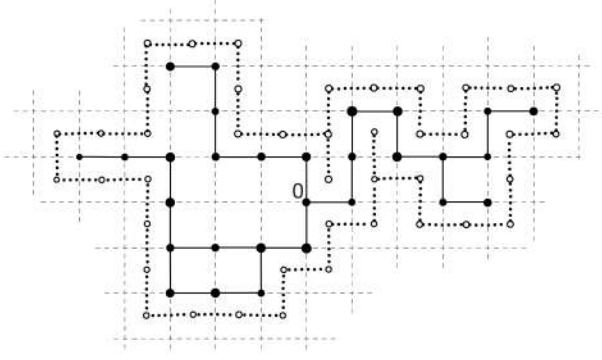


Figure 1: A finite open cluster at the origin, surrounded by an open circuit in the dual lattice.

$$\begin{aligned} \sum_{\gamma} \mathbb{P}_p(\gamma \text{ is open}) &\leq \sum_{n=1}^{\infty} (1-p)^n n \sigma(n-1) \leq 4 \sum_{n=1}^{\infty} 3^{n-2} (1-p)^n n \\ &= \frac{4}{9} \sum_{n=1}^{\infty} (3(1-p))^n n < \infty \quad \text{if } 3(1-p) < 1, \text{ that is, if } p > 2/3, \end{aligned}$$

where the summation is over all such γ . Furthermore, from the above inequalities we can see that

$$\sum_{\gamma} \mathbb{P}_p(\gamma \text{ is open}) \rightarrow 0 \quad \text{as } (1-p) \downarrow 0, \text{ or equivalently, as } p \uparrow 1$$

so we may find x satisfying $2/3 < x < 1$ such that

$$\sum_{\gamma} \mathbb{P}_p(\gamma \text{ is open}) \leq \frac{1}{2} \quad \text{if } p > x.$$

Finally, denoting the origin by o , by the union bound,

$$\begin{aligned} \theta(p) = \mathbb{P}_p(|C(o)| = \infty) &= \mathbb{P}_p(M(n) = 0 \text{ for all } n) = 1 - \mathbb{P}_p(M(n) \geq 1 \text{ for some } n) \\ &\geq 1 - \sum_{\gamma} \mathbb{P}_p(\gamma \text{ is open}) \geq \frac{1}{2} > 0 \quad \text{if } p > x, \end{aligned}$$

giving that $p_c(\mathbb{L}^2) \leq x < 1$. □

In fact, it can be shown that $p_c(\mathbb{L}^2) = 1/2$ exactly [Kes80], using the *FKG inequality*, which we study in Section 4. We do not show this here, however the proof can be found in [Gri99].

To conclude the proof of Theorem 3.2, it suffices to show that $p_c(\mathbb{L}^d) \leq p_c(\mathbb{L}^2)$ for all $d \geq 2$. Indeed, the d -dimensional lattice $\mathbb{L}^d$ may be embedded in $\mathbb{L}^{d+1}$ as the projection of $\mathbb{L}^{d+1}$ onto the subspace generated by the first d coordinates. With this embedding, the origin of $\mathbb{L}^{d+1}$ belongs to an infinite open cluster for a particular value of p whenever it belongs to an infinite open cluster of the sublattice $\mathbb{L}^d$. Thus $\theta(p) = \theta_{\mathbb{L}^d}(p)$ is non-decreasing in d , which implies that $p_c(\mathbb{L}^{d+1}) \leq p_c(\mathbb{L}^d)$ for $d \geq 1$, which completes the proof.

3.4 Computational Evidence

We now present computational evidence supporting these theoretical results. A standard method is to consider the probability that an infinite component exists, which we call $\psi(p)$. By the *zero-one law* for

occurrence of percolation in [PO], we have that for any fixed $p \in [0, 1]$, $\psi(p) \in \{0, 1\}$ almost surely and the threshold for the transition is indeed p_c .

Figure 2 displays Monte Carlo estimates of $\psi(p)$ obtained by simulating bond percolation on a 50×50 section of $\mathbb{L}^2$ over 300 independent trials for each value of p and finding the proportion of trials such that the configuration contains a crossing cluster in a fixed direction (left to right). The direction is chosen by convention, since the lattice being symmetric means that choosing the top-to-bottom direction yields equivalent results. Furthermore, this is an appropriate method of simulating an infinite component by a finite proxy, as is done in [Mal24].

Figure 3 shows example percolation configurations at three values of p , demonstrating the sharpness of the phase transition, even on a finite lattice. The simulation code can be found at [Bas26] and was written with the assistance of the Claude (Anthropic) Sonnet 4.5 generative AI model.

3.5 The Relationship Between Bond and Site Percolation and Their Thresholds

There are ways of impeding flow through a medium other than blocking the edges, and a natural alternative is to block the vertices instead [Gri99]. The corresponding model is termed ‘site percolation’.

Most arguments available for percolation models may be adapted to both bond and site models. Indeed, there is a sense in which every bond model may be reformulated as a site model (on a different graph); the converse is false, and therefore site models are more general than bond models [Gri99].

One may notice that vertex removal is more destructive to connectivity than edge removal, and thus expect that the critical probability is higher in the site percolation model compared to the bond percolation model. Given an arbitrary infinite connected graph G , it is natural to question the precise relationship between the two critical probabilities. In this pursuit, pick an origin vertex o and define p_c^{bond} as in Section 3.1. Define p_c^{site} analogously for a site percolation process on the same graph. We have the following theorem on their relationship [Gri99].

Theorem 3.6. *Let $G = (V, E)$ be an infinite connected graph with countably many edges, origin o , and maximum vertex degree Δ ($< \infty$). The critical probabilities of G satisfy*

$$\frac{1}{\Delta - 1} \leq p_c^{\text{bond}} \leq p_c^{\text{site}} \leq 1 - (1 - p_c^{\text{bond}})^\Delta. \quad (7)$$

One consequence of this theorem is that, if G is a graph with bounded degree, then $p_c^{\text{bond}}(G) < 1$ if and only if $p_c^{\text{site}}(G) < 1$.

4 The FKG Inequality

The FKG inequality, named after Fortuin, Kasteleyn, and Ginibre (1971) [Gri99], is a result stating that increasing events (as defined in Remark 2.4) are positively correlated. Such an inequality is useful in a variety of arguments in percolation theory; for example, in the proof of a uniqueness theorem which states that for $p > p_c$, the number of infinite clusters is 1 almost surely. In addition, as mentioned earlier, it is useful in the usual proof of the result that $p_c(\mathbb{L}^2) = 1/2$. For now, we simply state the theorem which is adapted from [Gri99], as it features prominently in later sections.

Theorem 4.1 (FKG inequality). *(a) If X and Y are increasing random variables such that $\mathbb{E}_p(X^2) < \infty$ and $\mathbb{E}_p(Y^2) < \infty$, then $\mathbb{E}_p(XY) \geq \mathbb{E}_p(X)\mathbb{E}_p(Y)$.*

(b) If A and B are increasing events then $\mathbb{P}_p(A \cap B) \geq \mathbb{P}_p(A)\mathbb{P}_p(B)$.

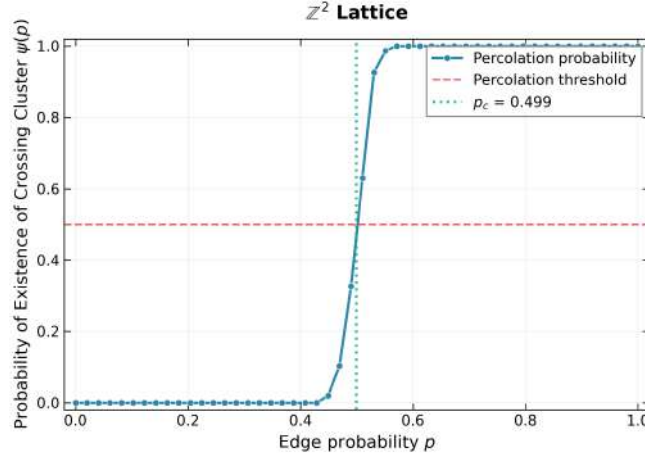


Figure 2: The probability of crossing cluster existence $\psi(p)$ as a function of p for bond percolation on a 50×50 section of $\mathbb{L}^2$. The phase transition occurs at $p_c \approx 0.499$, slightly different from 0.5 as expected due to the simulation and estimation being done on a finite lattice. On the infinite lattice, $\psi(p)$ jumps discontinuously from 0 to 1 at p_c whereas, on finite lattices, this jump is smoothed but remains centered at p_c , making the halfway point $\psi(p) = 1/2$ a natural estimator for p_c . As lattice size increases, the sigmoid transition sharpens and the estimate gets closer to the true critical probability.

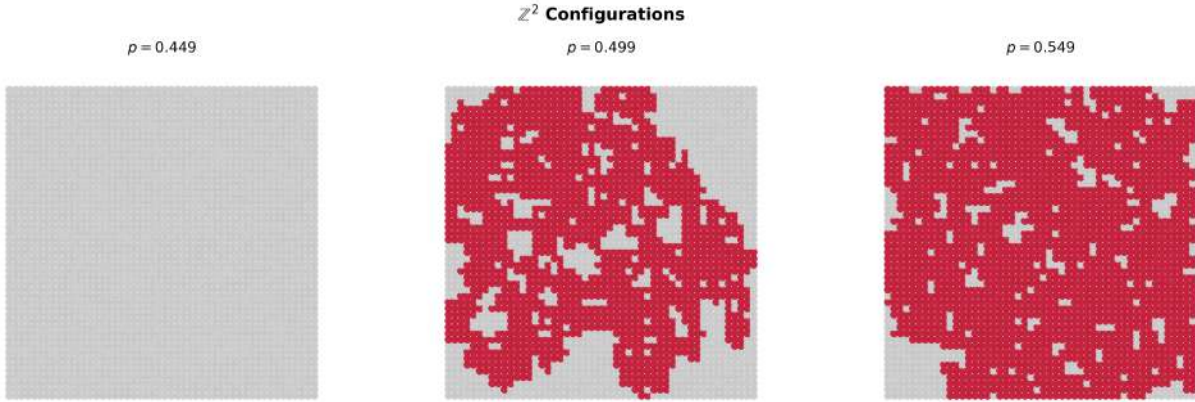


Figure 3: Bond percolation configurations on a 50×50 section of $\mathbb{L}^2$ for $p = p_c$ and two other values of p , one on each side of the threshold, indicating the swiftness of phase transition.

5 Simple Random Walks and Geometric Properties of Graphs

5.1 Electrical Networks

We now proceed to study several topics from [LP17] which prepare us for a discussion of [EST25]. In particular, we develop the theory which allows us to relate transience of random walks to the *isoperimetric profile* of a graph - a relationship exploited within the arguments in [EST25].

Definition 5.1. A network is a graph $G = (V, E)$ where each edge $e \in E$ is assigned a positive weight $c(e)$. Given a network $G = (V, E)$ with weights $c(\cdot)$ and some $K \subset V$, the induced network $G \upharpoonright K$ (denoting G restricted to K) is the subnetwork $G' = (K, E')$ where $E' = \{e = \{x, y\} \in E : x, y \in K\}$ with weights $c \upharpoonright E'$.

Definition 5.2. A random walk on a connected network $G = (V, E)$ is an irreducible, reversible Markov chain with states V and transition probabilities $p(x, y)$ between any two vertices $x, y \in V$ given by

$p(x, y) = \frac{c(x, y)}{\pi(x)}$ where $\pi(x)$ is the sum of the weights of the edges incident to x . If all edge weights are equal, we call the random walk a simple random walk. Note that $c(x, y) = c(y, x)$.

Definition 5.3. A function f is harmonic at a vertex x if

$$f(x) = \sum_{x \sim y} p(x, y) f(y).$$

Example 5.4. Suppose that A is a subset of states. Write τ_A for the first time the random walk visits some vertex in A ; if the chain happens to start in A , then $\tau_A = 0$. Consider the probability that the random walk visits A before it visits a disjoint subset Z as a function of its starting point x :

$$F(x) := \mathbb{P}_x(\tau_A < \tau_Z).$$

Conditioning on the first step, one can show that F is harmonic [LP17], and in the case of a simple random walk, we find that

$$F(x) := \frac{1}{\deg(x)} \sum_{x \sim y} F(y).$$

Definition 5.5. An electrical network is a network where we call the edge weights ‘conductances’ and their reciprocals ‘resistances’. We denote $c(x, y)^{-1}$ by $r(x, y)$.

Given two subsets A and Z of vertices of a network, a voltage function is a function on the vertices of the network which is harmonic at all $x \notin A \cup Z$.

Given a voltage function v , we define the associated ‘current’ function i on the edges by

$$i(x, y) := c(x, y)(v(x) - v(y)).$$

Remark 5.6. Consider now a connected electrical network and the probability that a random walk starting at a vertex a will hit a set Z before it returns to a . Write this as

$$\mathbb{P}(a \rightarrow Z) := \mathbb{P}_a(\tau_Z < \tau_a^+).$$

Consider a voltage v with value $v(a)$ at a and 0 on Z . The function $f(x) := \mathbb{P}_x(\tau_a < \tau_Z)$ is harmonic at all vertices in $V \setminus (\{a\} \cup Z)$ by Example 5.4, with $f(a) = 1$ and $f = 0$ on Z . Since f is harmonic, so is $v(a)f$, as scaling a harmonic function by a constant preserves harmonicity. Moreover, $v(a)f$ takes value $v(a)$ at a and 0 on Z . Since v is harmonic at all vertices in $V \setminus (\{a\} \cup Z)$ by definition, with the same values on $\{a\} \cup Z$, the uniqueness principle [LP17] gives $v(x) = v(a)f(x)$, thus $\mathbb{P}_x(\tau_a < \tau_Z) = v(x)/v(a)$. Then

$$\begin{aligned} \mathbb{P}(a \rightarrow Z) &= \sum_{x \in V} p(a, x) (1 - \mathbb{P}_x(\tau_a < \tau_Z)) = \sum_{x \in V} \frac{c(a, x)}{\pi(a)} \left(1 - \frac{v(x)}{v(a)}\right) \\ &= \frac{1}{v(a)\pi(a)} \sum_{x \in V} c(a, x)(v(a) - v(x)) = \frac{1}{v(a)\pi(a)} \sum_{x \in V} i(a, x). \end{aligned}$$

In other words,

$$v(a) = \frac{\sum_{x \in V} i(a, x)}{\pi(a)\mathbb{P}(a \rightarrow Z)}. \quad (8)$$

Since $\sum_{x \in V} i(a, x)$ is the *total amount of current flowing into the circuit at a* (by definition), we may regard the entire circuit between a and Z as a single conductor with *effective conductance* given by the following definition.

Definition 5.7. The effective conductance between a and Z is given by

$$\mathcal{C}_{\text{eff}} := \pi(a)\mathbb{P}(a \rightarrow Z) =: \mathcal{C}(a \leftrightarrow Z). \quad (9)$$

The effective resistance $\mathcal{R}(a \leftrightarrow Z)$ is the reciprocal of the effective conductance; in the case $a \in Z$, then we also define $\mathcal{R}(a \leftrightarrow Z) := 0$.

Remark 5.8. Note also that if there is unit current flowing from a to Z and the voltage on Z is 0, (8) and (9) imply that $v(a) = \mathcal{R}(a \leftrightarrow Z)$. Furthermore, Dirichlet's principle [LP17] implies the symmetry of effective resistance.

It is now natural to consider an extension of these notions of effective conductance and resistance to infinite networks. We approach this by taking large finite subgraphs. More precisely, for an infinite network G , let $(G_n)_{n \in \mathbb{N}}$ be any sequence of finite subgraphs of G that *exhaust* G , that is, $G_n \subseteq G_{n+1}$ and $G = \bigcup G_n$. Each edge in G_n is an edge in G , so we simply give it the same conductance it has in G . We also assume that G_n is the graph induced in G by $V(G_n)$. Let Z_n be the set of vertices in $G \setminus G_n$. Let G_n^W be the graph obtained from G by identifying Z_n to a single vertex, z_n , and then removing loops at z_n (but keeping multiple edges to Z_n). This graph will have finitely many vertices but may have infinitely many edges even when loops are deleted if some vertex of G_n has infinite degree. Given a network random walk on G , if we stop it the first time it reaches Z_n , then we obtain a network random walk on G_n^W until it reaches z_n . Now for every $a \in G$, the events $\{a \rightarrow Z_n\}$ are decreasing in n , so by downward continuity, the limit $\lim_{n \rightarrow \infty} \mathbb{P}(a \rightarrow Z_n)$ is the probability of never returning to a in G , which we call the *escape probability* from a . This is positive if and only if the random walk on G is transient. Hence, by (9), $\lim_{n \rightarrow \infty} \mathcal{C}(a \leftrightarrow z_n; G_n^W) > 0$ if and only if the random walk on G is transient. We call $\lim_{n \rightarrow \infty} \mathcal{C}(a \leftrightarrow z_n)$ the *effective conductance* from a to ∞ in G and denote it by $\mathcal{C}(a \leftrightarrow \infty)$. Its reciprocal, *effective resistance*, is denoted $\mathcal{R}(a \leftrightarrow \infty)$. This construction of effective conductance and resistance on infinite networks is called the *wired construction*; we collapse (wire) everything outside G_n to a single vertex ‘at infinity’, hence the superscript ‘W’. In particular, we have obtained:

Theorem 5.9. A random walk on an infinite connected network is transient iff the effective conductance from any of its vertices to infinity is positive.

Remark 5.10. In future, we may write $\mathcal{R}(A \leftrightarrow \infty)$ for some subset $A \subset V(G)$; by this we simply mean: identify A to a single vertex a , as above with Z_n to z_n , and consider instead $\mathcal{R}(a \leftrightarrow \infty)$, defined by the limit above.

Remark 5.11. $\mathcal{R}$ (the wired effective resistance) is a metric on an infinite connected network G . Indeed, it is clearly non-negative and the symmetry follows from Dirichlet's principle, as mentioned in Remark 5.8. Furthermore, by Definition 5.7, $\mathcal{R}(a \leftrightarrow a) = 0$ and if $a \neq x$, then $0 < \mathcal{C}(a \leftrightarrow x) < \infty$ so $\mathcal{R}(a \leftrightarrow x) > 0$. Finally, the triangle inequality holds in the finite network case by [LP17]. Hence, for any three vertices $x, y, z \in V(G)$, we can pick N large enough so that for all $n \geq N$, G_n contains x, y and z . Then we can invoke the finite-network triangle inequality and take the limit as $n \rightarrow \infty$, applying the algebra of limits to get the triangle inequality in the infinite network case: $\mathcal{R}(x \leftrightarrow y) \leq \mathcal{R}(x \leftrightarrow z) + \mathcal{R}(z \leftrightarrow y)$.

It will prove helpful later to note that we can replace y with ∞ . Indeed, before taking the limit as $n \rightarrow \infty$ above, simply replace y by z_n .

5.2 Isoperimetric Profiles

Define

$$\varphi(G, t) := \inf_{K \subset V} \{|\partial_E K|_c; t \leq |K|_\pi < \infty\},$$

where G is a network, K ranges over finite sets of vertices in G , $|K|_\pi$ is the π -measure of K (i.e. $|K|_\pi := \sum_{v \in K} \pi(v)$), $t > 0$ is a lower threshold on $|K|_\pi$, $\partial_E K$ denotes the edge boundary of K (the set of

edges with exactly one endpoint in K), and $|\partial_E K|_c$ is the total conductance of this edge boundary (i.e. $|\partial_E K|_c := \sum_{e \in \partial_E K} c(e)$).

We call $\varphi(G, t)$ an *isoperimetric profile* of G . It captures how well-connected the graph is globally: a large profile means every large set has a large boundary, while a small profile indicates bottlenecks where large regions can be cut off cheaply.

Example 5.12. If $G = \mathbb{L}$, $\varphi(G, t) = 2$ (taking all edge weights to be 1) for all t since the boundary of any connected subset is always 2 - one edge at either end. Note that disconnected sets will never be minimisers.

For applications to effective resistance, however, we need a more refined version that restricts attention to sets growing outward from a fixed anchor A , leading to the *anchored* isoperimetric profile $\varphi(G, A, t)$ given by

$$\varphi(G, A, t) := \inf_{K \subset V} \{ |\partial_E K|_c; A \subseteq K, K/A \text{ is connected}, t \leq |K|_\pi < \infty \} \quad (10)$$

for $A \subset V(G)$. Here, we say that K/A is connected when the graph induced by K in G/A is connected, where we have identified all of A to a single vertex. The requirement that K contain A and K/A be connected is what leads to the adjective ‘anchored.’ The following definition ‘summarises’ the isoperimetric profile, essentially asking which power law best describes its growth in t .

Definition 5.13. *The isoperimetric dimension of a graph $G = (V, E)$ is given by*

$$\text{Dim}(G) := \sup \left\{ d \geq 1 : \inf_{\substack{K \subset V \\ 0 < |K|_\pi < \infty}} \frac{|\partial_E K|_c}{|K|_\pi^{(d-1)/d}} > 0 \right\}. \quad (11)$$

Remark 5.14. If $\text{Dim}(G) > d'$ for some $d' \in \mathbb{R}$, then by properties of the supremum, there exists $d > d'$ in the above set over which the supremum is taken. Thus there exists $\beta > 0$ such that $|\partial_E K|_c \geq \beta |K|_\pi^{(d-1)/d}$ for all finite K . Since the anchored profile takes an infimum over a smaller class of sets than the unanchored profile, we immediately get $\varphi(G, A, t) \geq \varphi(G, t) \geq \beta t^{(d-1)/d}$ for all $t > 0$. Note that this bound is uniform - it does not depend on the anchor set A .

5.3 Random Walk Transience

We have now laid the groundwork that allows us to state and prove a theorem which, given a certain *isoperimetric inequality* (a bound on the anchored isoperimetric profile), bounds from above the effective resistance from a finite set to infinity. In turn, this allows us to infer a sufficient condition on the isoperimetric dimension of the graph for uniform transience of the simple random walk. Here, it is important to remember that a graph is an unweighted network, and so the random walk will be simple. We tell the story first, before returning and completing the proofs of the results used. The first (and most important) theorem is Theorem 6.41 from [LP17].

Theorem 5.15. *Let A be a finite set of vertices in a network G with $|V(G)|_\pi = \infty$. Suppose that $\varphi(t) := \varphi(G, A, t)$ has the property that $\varphi(t) \geq f(t)$ for some increasing function f on $[|A|_\pi, \infty)$ that satisfies $0 < f(t) \leq t$ and $f(2t) \leq \alpha f(t)$ for some α . Then*

$$\mathcal{R}(A \leftrightarrow \infty) \leq \int_{|A|_\pi}^{\infty} \frac{4\alpha^2}{f(t)^2} dt.$$

Recall from above that for a vertex $a \in V(G)$ where G is an infinite network, $\mathcal{C}(a \leftrightarrow \infty) > 0$ if and only if the random walk starting at a is transient. This is equivalent to $\mathcal{R}(a \leftrightarrow \infty) < \infty$. Moreover, since $\mathcal{R}(a \leftrightarrow \infty) = (\pi(a)\mathbb{P}(a \rightarrow \infty))^{-1}$, a uniform upper bound $\mathcal{R}(a \leftrightarrow \infty) \leq C < \infty$ over all $a \in V(G)$

gives $\pi(a)\mathbb{P}(a \rightarrow \infty) \geq 1/C > 0$ uniformly, which is precisely uniform transience. Thus, if we can find f satisfying the conditions of Theorem 5.15 such that the integral is finite and independent of a , the bound will hold uniformly over all a and the random walk will be uniformly transient.

In pursuit of this, let G be an infinite, connected graph. Then $|V(G)|_\pi = \infty$ since the total edge weights out of each vertex are all positive (by connectivity), and there are infinitely many vertices. Furthermore, if $d' < \dim(G)$ for some fixed $d' \in \mathbb{R}$ then Remark 5.14 implies that we can pick $0 < \beta < 1$ and $d > d'$ such that taking $f(t) := \beta t^{(d-1)/d}$ satisfies the isoperimetric inequality in Theorem 5.15, since f is clearly increasing in t since $(d-1)/d \geq 0$. Note that for any vertex $a \in A$, $t \geq |\{a\}|_\pi \geq 1$ so $t^{(d-1)/d} \leq t$. Since $\beta < 1$, we have that $0 < f(t) \leq t$, satisfying the second inequality in the theorem. Now, $f(2t) = 2^{(d-1)/d} f(t)$, so we may take $\alpha = 2^{(d-1)/d}$, satisfying the final inequality.

Theorem 5.15 therefore applies and gives, for all vertices a , that

$$\mathcal{R}(a \leftrightarrow \infty) \leq \int_1^\infty \frac{4 \cdot 2^{2(d-1)/d}}{\beta^2 t^{2(d-1)/d}} dt.$$

This integral is indeed independent of a and converges if and only if $\frac{2(d-1)}{d} > 1$, that is, $d > 2$. Therefore, taking $d' = 2$ means that $d > 2$, and we have proved the following theorem.

Theorem 5.16. *If G is an infinite, connected graph and $\dim(G) > 2$, then G is uniformly transient.*

We now proceed to prove Theorem 5.15 through a convolution of several results and a chain of implications, the first of which we now state.

Theorem 5.17. *Let A be a finite set of vertices in a network G with $|V(G)|_\pi = \infty$. Let $\varphi(t) := \varphi(G, A, t)$. Define $s_0 := |A|_\pi$ and $s_{k+1} := s_k + \varphi(s_k)/2$ recursively for $k \geq 0$. Then*

$$\mathcal{R}(A \leftrightarrow \infty) \leq \sum_{k \geq 0} \frac{2}{\varphi(s_k)}.$$

Before proving Theorem 5.17, we show how it implies Theorem 5.15.

Proof of Theorem 5.15. Define $t_0 := |A|_\pi$ and $t_{k+1} := t_k + f(t_k)/2$ recursively. We have that $s_k \geq t_k$ and $t_k \leq t_{k+1} \leq 2t_k$ by the conditions of the theorem. Whence, similarly, for $t_k \leq t \leq t_{k+1}$, we have $f(t) \leq f(2t_k) \leq \alpha f(t_k)$, so that

$$\begin{aligned} \int_{|A|_\pi}^\infty \frac{4\alpha^2}{f(t)^2} dt &\geq \sum_{k \geq 0} \int_{t_k}^{t_{k+1}} \frac{4\alpha^2}{f(t)^2} dt \geq \sum_{k \geq 0} \int_{t_k}^{t_{k+1}} \frac{4}{f(t_k)^2} dt = \sum_{k \geq 0} \frac{4(t_{k+1} - t_k)}{f(t_k)^2} \\ &= \sum_{k \geq 0} \frac{2f(t_k)}{f(t_k)^2} \geq \sum_{k \geq 0} \frac{2}{\varphi(t_k)} \geq \sum_{k \geq 0} \frac{2}{\varphi(s_k)} \geq \mathcal{R}(A \leftrightarrow \infty) \end{aligned}$$

where the penultimate inequality follows from φ being non-decreasing by definition, and the final inequality is precisely Theorem 5.17. $\square$

We show Theorem 5.17 by means of the following two lemmas - we state an analogue for finite networks, and then its extension to the infinite case. For the purposes of brevity and space constraints, while the implications between each will be shown, these lemmas will be stated but not proved. Their proofs can be found in [LP17] as results (6.43) and (6.18) respectively.

Lemma 5.18. *Let a and z be two distinct vertices in a finite connected network G . Define*

$$\phi(t) := \phi(G, t) := \min_{W \subset V} \{|\partial_E W|_c; a \in W, z \notin W, W \text{ is connected}, t \leq |W|_\pi\}$$

for $t \leq |V(G) \setminus \{z\}|_\pi$ and $\phi(t) := \infty$ for $t > |V(G) \setminus \{z\}|_\pi$. Define $s'_0 := \pi(a)$ and $s'_{k+1} := s'_k + \phi(s'_k)/2$ recursively for $k \geq 0$. Then

$$\mathcal{R}(a \leftrightarrow z) \leq \sum_{k \geq 0} \frac{2}{\phi(s'_k)}.$$

Lemma 5.19. *Let A be a finite set of vertices in a connected network G with $|V(G)|_\pi = \infty$. Let H be a finite connected subnetwork (a subgraph which is a network) containing A . Call H' the graph H , where we identify A to a single vertex a , and the set of boundary vertices of H (denoted ∂H) to a single vertex z . Define $\varphi(G, A, t)$ and $(s_k)_{k \geq 0}$ as in Theorem 5.17 and $\phi(H', t)$ with its associated sequence $(s'_k)_{k \geq 0}$ as in Lemma 5.18. Then for all $t \geq 0$,*

$$\phi(H', \pi(a) + t) \geq \varphi(G, A, |A|_\pi + t),$$

and for all $k \geq 0$, we have $\phi(H', s'_k) \geq \varphi(G, A, s_k)$.

Proof of Theorem 5.17. Consider again the setting of the theorem. Applying Lemma 5.18 to H' as in Lemma 5.19 gives

$$\mathcal{R}(a \leftrightarrow z; H') \leq \sum_{k \geq 0} \frac{2}{\phi(H', s'_k)},$$

and by Lemma 5.19 we have $\phi(H', s'_k) \geq \varphi(G, A, s_k)$ for all $k \geq 0$, so

$$\mathcal{R}(a \leftrightarrow z; H') \leq \sum_{k \geq 0} \frac{2}{\varphi(G, A, s_k)}.$$

Since this bound is independent of the choice of subnetwork H containing A , it holds for every element of any exhaustion $(H_n)_{n \in \mathbb{N}}$ of G by finite connected subnetworks containing A . Also, by definition (5.10), $\mathcal{R}(A \leftrightarrow \infty) = \lim_{n \rightarrow \infty} \mathcal{R}(a \leftrightarrow z; H'_n)$ so

$$\mathcal{R}(A \leftrightarrow \infty) \leq \sum_{k \geq 0} \frac{2}{\varphi(s_k)},$$

which is precisely the statement of Theorem 5.17. □

6 Counting Minimal Cutsets and $p_c < 1$

This section is devoted to reviewing and understanding [EST25] - a paper by Easo, Severo and Tassion, which proves two results concerning percolation on general graphs.

6.1 Main Results

Definition 6.1. *Let $G = (V, E)$ be an infinite, connected, locally finite graph. A set of edges $F \subset E$ is called a cutset from a vertex v to ∞ if v belongs to a finite connected component of $(V, E \setminus F)$. A cutset is called minimal if no proper subset of it is a cutset.*

Let $Q_n(v)$ be the set of minimal cutsets from v to ∞ of cardinality n and consider the quantity

$$q_n := \sup_{v \in V} |Q_n(v)|.$$

Here $|\emptyset| := 0$, and we note that $q_n = \infty$ is possible, for example for $G = \mathbb{L}$ and $n = 2$. The authors are interested in cases where the number of cutsets q_n grows at most exponentially with n , and define

$$\kappa(G) := \sup_{n \geq 1} q_n^{1/n}.$$

Definition 6.2. *The critical parameter for ‘uniform’ percolation is*

$$p_c^*(G) := \inf\{p \in [0, 1] : \theta^*(p) > 0\},$$

where $\theta^*(p) := \inf_{v \in V} \theta_v(p)$.

By the classical Peierls argument [Pei36], if $\kappa(G) < \infty$, then percolation on G has a uniformly percolating phase in the sense that $p_c^*(G) < 1$ [EST25]. The first result of the paper is a converse; they prove that if $p_c^*(G) < 1$, the number of minimal cutsets of size n separating a given vertex from infinity is bounded above exponentially in n . Moreover, they prove that $p_c^*(G) < 1$ for every uniformly transient graph. More precisely, they state the following theorems.

Theorem 6.3 ([EST25] Theorem 1). *For every infinite, connected, locally finite graph G we have*

$$p_c^*(G) < 1 \iff \kappa(G) < \infty.$$

Theorem 6.4 ([EST25] Theorem 2). *Let G be an infinite, connected, locally finite graph. If G is uniformly transient, then $\kappa(G) < \infty$.*

Remark 6.5. Theorem 6.4 implies from Theorem 6.3 that uniform transience of an infinite, connected, locally-finite graph G implies $p_c^*(G) < 1$. Theorem 5.16, the main theorem in Section 5, therefore implies that $p_c^*(G) < 1$ for every such graph G satisfying $\text{Dim}(G) > 2$.

Remark 6.6. The uniform critical parameter p_c^* differs slightly from the critical parameter defined in Section 3.1, in that the uniform version requires the percolation probability to be bounded away from zero over all vertices, not just that they are all positive. However, these notions often coincide, such as for graphs that are (quasi-) transitive [EST25].

Definition 6.7. *A graph $G = (V, E)$ is transitive (or vertex-transitive) if its automorphism group acts transitively on vertices. That is, for every pair $x, y \in V$ there exists an automorphism (a bijection from the vertices to themselves preserving edge structure) of the graph G that maps x to y .*

This definition is equivalent to asking that the group of automorphisms of G has only one orbit (a set of elements reachable from a specific vertex under group actions). A slight generalisation can be made through the following definition.

Definition 6.8. *G is quasi-transitive if the group of automorphisms on the vertex set has only finitely many orbits.*

While the paper [EST25] presents many interesting results as corollaries of the two main theorems above, whether new or improved, there are too many to explore within the space constraints of this report. For this reason, we mention only one more particularly interesting conclusion.

Remark 6.9. The authors establish the existence of a universal constant $\varepsilon > 0$ such that every infinite, connected, locally finite, transitive graph G satisfies either $p_c = 1$ or $p_c \leq 1 - \varepsilon$. This follows from the proof of Theorem 1.7 in [HT24] if G is recurrent [EST25]. If G is transient, it follows from the proof of Theorem 6.4, which we shall see shortly.

6.2 Proof of Theorem 6.3 (Theorem 1)

Let $G = (V, E)$ be an infinite, connected, locally finite graph. The if-and-only-if statement in Theorem 6.3 is equivalent to:

$$(\exists p < 1 \ \exists \theta > 0 \ \forall u \in V \ \mathbb{P}_p(u \leftrightarrow \infty) \geq \theta) \iff (\exists K < \infty \ \forall u \in V \ \forall n \geq 1 \ |Q_n(u)| \leq K^n)$$

Proof. Within this proof, we state and use results which we prove afterwards.

The implication $\Leftarrow$ follows precisely from the Peierls argument, which we saw in Section 3.3, generalised to the graph G [EST25]. We sketch it now for completeness.

Lemma 6.10. *Let $S \subset V$ be a finite connected set. For every $u \in S$, $\partial_\infty S$ is a minimal cutset from u to ∞ .*

If the cluster of some $u \in V$ is finite, the above lemma gives that its exposed boundary is a finite minimal cutset from u to ∞ with all edges being closed. As in Section 3.3, the union bound gives that, for every $p \in [0, 1]$,

$$\mathbb{P}_p(|C(u)| < \infty) \leq \sum_{n \geq 1} q_n(1-p)^n. \quad (12)$$

If $q_n \leq K^n$ for some constant $K < \infty$, then the right hand side above converges to 0 as p tends to 1. Identically again to Section 3.3, since the bound is uniform in u , there exists $p < 1$ such that

$$\forall u \in V \quad \mathbb{P}_p(u \leftrightarrow \infty) \geq 1/2.$$

It remains to prove the implication $\Rightarrow$, which we do now. Fix $\theta, p \in (0, 1)$ such that $\mathbb{P}_p(u \leftrightarrow \infty) \geq \theta$ for every $u \in V$. Fix $o \in V$ and $n \geq 1$. Writing C for the cluster of o , we show that for every minimal cutset Π from o to ∞ with $|\Pi| = n$,

$$\mathbb{P}_p(\partial_\infty C = \Pi) \geq 1/K^n, \quad (13)$$

where $K = K(p, \theta) \in (0, \infty)$ is a finite constant depending on p and θ only (in particular it does not depend on the chosen vertex o), and $\partial_\infty C$ is the *exposed boundary* of C - the set of edges $\{u, v\} \in E$ such that $u \in C$ and v is connected to ∞ in $V \setminus C$.

This concludes the proof since

$$1 \geq \sum_{\Pi \in Q_n(o)} \mathbb{P}_p(\partial_\infty C = \Pi) \geq |Q_n(o)|/K^n.$$

where the first inequality is due to the sum being over mutually exclusive events, and the second inequality is exactly (13).

Lemma 6.11. *Let $o \in V$, let Π be a minimal cutset from o to ∞ . Let A be the connected component of o in $(V, E \setminus \Pi)$ and $B = \{e \cap A : e \in \Pi\}$ be the set of inner vertices of Π . For every set S of vertices, we have*

$$(B \subset S \subset A) \implies (\partial_\infty S = \Pi).$$

Let A, B and Π be as in Lemma 6.11. Any infinite open path from a vertex $u \in A$ must intersect B before exiting A , thus the hypothesis $\mathbb{P}_p(u \leftrightarrow \infty) \geq \theta$ implies that for all $u \in A$, $\mathbb{P}_p(u \overset{A}{\leftrightarrow} B) \geq \theta$, where this notation denotes the probability that u can reach B through an open path with vertices in A .

Proposition 6.12. *Let $G = (V, E)$ be a finite, connected graph. Let $B \subset V$, let $\theta, p \in (0, 1]$, and suppose that $\mathbb{P}_p(u \leftrightarrow B) \geq \theta$ for every $u \in V$. Define $c := \left(\frac{p\theta}{2}\right)^{3/\theta}$. Then for every $o \in V$,*

$$\mathbb{P}_p\left(\bigcap_{b \in B} \{o \leftrightarrow b\}\right) \geq c^{|B|}.$$

Let $\mathcal{E}$ be the event that every vertex in B is connected to o by an open path in A . By Proposition 6.12 applied to the finite subgraph of G induced by A , we have $\mathbb{P}_p(\mathcal{E}) \geq c^n$ where $c = (p\theta/2)^{3/\theta} > 0$, since $|B| \leq n$ and $c < 1$ so $c^{|B|} \geq c^n$.

Let $\mathcal{F}$ be the event that all the edges of Π are closed. Note that Π is deterministic and up until this point, admits no configuration of open or closed edges. Furthermore, $\mathcal{E}$ depends only on edges within A . Thus, $\mathcal{E}$ and $\mathcal{F}$ are independent, and

$$\mathbb{P}_p(\mathcal{E} \cap \mathcal{F}) = \mathbb{P}_p(\mathcal{E})\mathbb{P}_p(\mathcal{F}) \geq c^n(1-p)^n.$$

If the event $\mathcal{E} \cap \mathcal{F}$ occurs, then the cluster C of o satisfies $B \subset C \subset A$. Indeed, $B \subset C$ by $\mathcal{E}$, and any path exiting A must cross at least one edge of Π , but no such path exists by $\mathcal{F}$, so $C \subset A$. Hence, by Lemma 6.11 we must have $\partial_\infty C = \Pi$. This concludes that

$$\mathbb{P}_p(\partial_\infty C = \Pi) \geq \mathbb{P}_p(\mathcal{E} \cap \mathcal{F}) \geq c^n(1-p)^n,$$

which establishes (13) with $K = \frac{1}{c(1-p)} = \frac{1}{(p\theta/2)^{3/\theta}(1-p)}$. $\square$

Proof of Lemma 6.10. Any path from u to ∞ in V must traverse an edge in $\partial_\infty S$ (consider the last edge traversed by this path intersecting S). Therefore, $\partial_\infty S$ is a cutset from u to ∞ . To prove that it is minimal, consider an edge $e \in \partial_\infty S$. Since S is connected, there exists a path from u to an endpoint of e in S and by definition of the exposed boundary, there must exist a path from the other endpoint of e to ∞ in $V \setminus S$. The concatenation of these two paths with e connects u to ∞ without using any edges of $\partial_\infty S$ other than e . Hence $\partial_\infty S \setminus \{e\}$ is not a cutset from u to ∞ . $\square$

Proof of Lemma 6.11. Since A is a maximal connected set in $(V, E \setminus \Pi)$, all the edges at the boundary of A belong to Π . Indeed, if an edge $e = \{x, y\}$ was such that $e \in \partial A$ with $x \in A$ and $y \notin A$ but $e \notin \Pi$, then it connects x to y in $(V, E \setminus \Pi)$, but this means that o is connected to y in that set, contradicting that $y \notin A$. Therefore $\partial_\infty A \subset \partial A \subset \Pi$.

By Lemma 6.10, $\partial_\infty A$ is a cutset from o to ∞ , so by the minimality of Π , the two inclusions above must be equalities:

$$\partial_\infty A = \partial A = \Pi.$$

Now, let S be a set satisfying $B \subset S \subset A$; we prove the implication equality by showing both inclusions. Firstly, let $e \in \Pi$. Since $e \in \partial_\infty A$, one endpoint of e must belong to B (by definition) and the other endpoint is connected to ∞ in $V \setminus A$. Therefore, by hypothesis, one endpoint of e belongs to S and the other endpoint is connected to ∞ in $V \setminus S$. This proves the inclusion $\Pi \subset \partial_\infty S$.

Now let $e \in \partial_\infty S$. Let u be the endpoint of e in S , and let v be the endpoint of e connected to ∞ in $V \setminus S$. Then, by hypothesis, $u \in A$ and v is connected to ∞ in $V \setminus B$. Since $\Pi = \partial A$, every edge in Π intersects A and thus B , so there must exist an infinite path starting at v in the subgraph $(V, E \setminus \Pi)$. In particular, $v \notin A$, and hence $e \in \partial A = \Pi$. Hence $\partial_\infty S \subset \Pi$. $\square$

Proof of Proposition 6.12. Say that a finite sequence of vertices $x_1, \dots, x_k$ is *chained* if $x_1 = o$ and for all $i \in \{2, \dots, k\}$,

$$\frac{p\theta}{2} \leq \mathbb{P}_p(x_i \leftrightarrow \{x_1, \dots, x_{i-1}\}) \leq \frac{\theta}{2}. \quad (\mathbf{P1})$$

Since there exists at least one chained sequence (take $k = 1$, then the statement is vacuously true) and V is finite, there must exist a chained sequence $x_1, \dots, x_k$ that is *maximal* in the sense that for every vertex x_{k+1} , the sequence $x_1, \dots, x_{k+1}$ is not chained. Fix a maximal chained sequence $x_1, \dots, x_k$, and let

$X := \{x_1, \dots, x_k\}$. We claim that, in addition to **(P1)**, this sequence satisfies the following two properties, where $n := |B|$:

$$\forall u \in V \quad \mathbb{P}_p(u \leftrightarrow X) \geq \frac{\theta}{2}, \quad (\mathbf{P2})$$

$$k \leq \frac{2n}{\theta}. \quad (\mathbf{P3})$$

To prove **(P2)**, consider the set of vertices $W \subset V$ that are connected to X with probability at least $\theta/2$ and suppose for contradiction that $W \neq V$. Since W is non empty (because $X \subset V$) and G is connected, we can consider an edge $\{u, v\}$ such that $u \in W$ and $v \notin W$. By the FKG inequality - Theorem 4.1 part (b) - we have that

$$\theta/2 > \mathbb{P}_p(v \leftrightarrow X) \geq \mathbb{P}_p(\{u, v\} \text{ is open}) \cdot \mathbb{P}_p(u \leftrightarrow X) \geq \frac{p\theta}{2}.$$

In particular, $x_1, \dots, x_k, v$ is a chained sequence, contradicting the maximality of $x_1, \dots, x_k$.

We now prove **(P3)**. To this aim, for each $i \in \{1, \dots, k\}$, let N_i denote the number of clusters that intersect both $\{x_1, \dots, x_i\}$ and B . For every $i \in \{2, \dots, k\}$, the increment $N_i - N_{i-1}$ is equal to 1 if x_i is connected to B but not to the previous points $\{x_1, \dots, x_{i-1}\}$, and it is equal to 0 otherwise. Therefore, for every $i \in \{2, \dots, k\}$, we have the deterministic inequality

$$N_i - N_{i-1} \geq \mathbf{1}_{x_i \leftrightarrow B} - \mathbf{1}_{x_i \leftrightarrow \{x_1, \dots, x_{i-1}\}}.$$

Taking the expectation, using our hypothesis and **(P1)**, for every $i \in \{2, \dots, k\}$, we get

$$\mathbb{E}_p(N_i) - \mathbb{E}_p(N_{i-1}) \geq \underbrace{\mathbb{P}_p(x_i \leftrightarrow B)}_{\geq \theta} - \underbrace{\mathbb{P}_p(x_i \leftrightarrow \{x_1, \dots, x_{i-1}\})}_{\leq \theta/2} \geq \theta/2.$$

Summing over $i \in \{2, \dots, k\}$ and using $\mathbb{E}_p(N_1) = \mathbb{P}_p(x_1 \leftrightarrow B) \geq \theta \geq \theta/2$, we get $\mathbb{E}_p[N_k] \geq \frac{\theta}{2}k$. Since N_k is deterministically bounded above by $|B| = n$, this concludes the proof of **(P3)**.

We now explain how the three properties above of the chained sequence imply the desired lower bound in the proposition. First, we estimate the event that all the vertices of X are connected to o : by a sequential reformulation, the FKG inequality, **(P1)** and **(P3)**, we have

$$\begin{aligned} \mathbb{P}_p \left(\bigcap_{u \in X} \{o \leftrightarrow u\} \right) &\geq \mathbb{P}_p \left(\bigcap_{i=2}^k \{x_i \leftrightarrow \{x_1, \dots, x_{i-1}\}\} \right) \\ &\geq \prod_{i=2}^k \mathbb{P}_p(x_i \leftrightarrow \{x_1, \dots, x_{i-1}\}) \geq \left(\frac{p\theta}{2} \right)^{k-1} \geq \left(\frac{p\theta}{2} \right)^{\frac{2n}{\theta}}. \end{aligned}$$

Second, we estimate the event that all the vertices of B are connected to X : by **(P2)** and the FKG inequality again, we have

$$\mathbb{P}_p \left(\bigcap_{b \in B} \{b \leftrightarrow X\} \right) \geq \prod_{b \in B} \mathbb{P}_p(b \leftrightarrow X) \geq \left(\frac{\theta}{2} \right)^n.$$

If all the vertices of X are connected to o and all the vertices of B are connected to X , then all the vertices of B are connected to o . Hence, by the two displayed equations above, the aforementioned inclusion and the FKG inequality once more, we obtain

$$\mathbb{P}_p \left(\bigcap_{b \in B} \{o \leftrightarrow b\} \right) \geq \mathbb{P}_p \left(\bigcap_{u \in X} \{o \leftrightarrow u\} \right) \cdot \mathbb{P}_p \left(\bigcap_{b \in B} \{b \leftrightarrow X\} \right) \geq \left(\frac{p\theta}{2} \right)^{\frac{2n}{\theta}} \cdot \left(\frac{\theta}{2} \right)^n \geq c^n,$$

where $c := \left(\frac{p\theta}{2}\right)^{3/\theta}$. □

6.3 Proof of Theorem 6.4 (Theorem 2)

Recall that we wish to prove that uniform transience of the random walk on an infinite, connected, locally finite graph G implies $\kappa(G) < \infty$.

While uniform transience was mentioned earlier, we define it properly here; we say that a graph G is uniformly transient if there exists $\varepsilon > 0$ such that $\mathcal{C}(v \leftrightarrow \infty) \geq \varepsilon$ for every vertex $v \in V$. That is, the simple random walk $(X_t)_{t=0}^\infty$ on G started at v satisfies

$$\mathcal{C}(v \leftrightarrow \infty) = d_v \mathbb{P}_v(\forall t \geq 1 : X_t \neq v) \geq \varepsilon.$$

Proof. Let G be uniformly transient as defined above. Let $G' = (V', E')$ be the graph obtained from G by replacing each edge by a path of length 2. View V as a subset of V' , and let $m : E \rightarrow V'$ map each edge to its midpoint. Let $\mathbb{P}'_u$ be the law of the simple random walk in G' started from a given vertex u , and let $\tau := \sup\{t \geq 0 : X'_t = X'_0\}$. We claim that for all $z \in V'$,

$$d_z \mathbb{P}'_z(\tau = 0) \geq \varepsilon_1 := \frac{2\varepsilon}{4 + \varepsilon}. \quad (14)$$

Note that if $z \in V$, the simple random walk on G' induces a *lazy* simple random walk on G - a random walk where there is a fixed probability 1/2 of staying at a vertex after a step, and probability 1/2 of leaving, the direction being chosen uniformly over adjacent vertices. This is because the first step will be to an edge midpoint, and the second step may be back to z , effectively remaining at z from the perspective of G . In this case, (14) follows by uniform transience of G . Indeed, let $s_0 = 0$, $s_{k+1} = \inf\{t > s_k : X'_t \in V\}$, and set $Y_k = X'_{s_k}$. By the strong Markov property, $(Y_k)_{k \geq 0}$ is a lazy simple random walk on G started at z . Then

$$\begin{aligned} d_z \mathbb{P}'_z(\tau = 0) &= d_z \mathbb{P}'_z(X'_t \neq z \ \forall t \geq 1) = d_z \mathbb{P}_z^{\text{lazy}}(Y_k \neq z \ \forall k \geq 1) = \frac{1}{2} \sum_{v \sim z} \mathbb{P}_v^{\text{lazy}}(Y_k \neq z \ \forall k \geq 0) \\ &= \frac{1}{2} \sum_{v \sim z} \mathbb{P}_v(X_t \neq z \ \forall t \geq 0) = \frac{d_z}{2} \mathbb{P}_z(X_t \neq z \ \forall t \geq 1) \geq \frac{\varepsilon}{2} > \varepsilon_1 \end{aligned} \quad (15)$$

where the third equality is by the Markov property and one-step decomposition of the lazy simple random walk on the first step, and the fifth equality is the reverse.

Otherwise, let $z = m(\{u, v\})$ for some $\{u, v\} \in E$, and let $\ell_x := |\{t \geq 0 : X'_t = x\}|$. Then

$$\frac{1}{\mathbb{P}'_z(\tau = 0)} = \sum_{n \geq 0} \mathbb{P}'_z(\tau > 0)^n = \mathbb{E}'_z[\ell_z] = \sum_{t \geq 0} \mathbb{P}'_z(X'_t = z)$$

where the first equality follows from applying the geometric series formula and noting that uniform transience of G implies transience of G' , justifying its application. The second equality follows simply from the tail sum formula for the expectation of a non-negative, integer-valued random variable, with implicit use of the strong Markov property. The third equality follows from writing ℓ_z as a sum of indicators and using the linearity of expectation. Now, by the law of total probability and since z is an edge midpoint so has only two neighbours,

$$\sum_{t \geq 0} \mathbb{P}'_z(X'_t = z) = 1 + \sum_{t \geq 1} \left[\mathbb{P}'_z(X'_{t-1} = u) \cdot \frac{1}{d_u} + \mathbb{P}'_z(X'_{t-1} = v) \cdot \frac{1}{d_v} \right]$$

$$\begin{aligned}
&= 1 + \frac{\mathbb{E}'_z[\ell_u]}{d_u} + \frac{\mathbb{E}'_z[\ell_v]}{d_v} \\
&\leq 1 + \frac{\mathbb{E}'_u[\ell_u]}{d_u} + \frac{\mathbb{E}'_v[\ell_v]}{d_v} = 1 + \frac{1}{d_u \mathbb{P}'_u(\tau = 0)} + \frac{1}{d_v \mathbb{P}'_v(\tau = 0)},
\end{aligned}$$

where the inequality follows from the fact that $\mathbb{E}'_z[\ell_u] = \mathbb{P}'_z(\text{reach } u) \mathbb{E}'_u[\ell_u]$ and the strong Markov property again. Now, by (15),

$$d_z \mathbb{P}'_z(\tau = 0) \geq d_z \left[1 + \frac{1}{d_u \mathbb{P}'_u(\tau = 0)} + \frac{1}{d_v \mathbb{P}'_v(\tau = 0)} \right]^{-1} \geq 2 \left[1 + \frac{4}{\varepsilon} \right]^{-1} = \varepsilon_1$$

and so we have proved (14).

Let $C := \{X'_t : 0 \leq t \leq \tau\}$ and $\partial := \{e \cap C : e \in \partial_\infty C\}$. Fix $o \in V$, and pick a neighbour $o' \in m(E)$ of o in G' . Fix a finite minimal cutset Π from o to ∞ in G , and set $n := |\Pi|$. We will show that for some finite constant $K \in (0, \infty)$ depending only on ε ,

$$\mathbb{P}'_{o'}(\partial = m(\Pi)) \geq 1/K^n. \quad (16)$$

This implies that $\kappa(G) < \infty$ because for all $o \in V$ and $n \geq 1$, by (16),

$$1 \geq \sum_{\Pi \in Q_n(o)} \mathbb{P}'_{o'}(\partial = m(\Pi)) \geq |Q_n(o)|/K^n.$$

Let A be the connected component of o in $(V, E \setminus \Pi)$. Let I be the connected component of o in $(V', E' \setminus \Pi')$, where Π' is the set of edges of G' incident to $m(\Pi)$. Note that $A \subset I$. Let $U := m(\Pi) \cup \{o'\}$ so that if the simple random walk visits every vertex of U and returns to o' before leaving $U \cup I$, we have that $\partial = m(\Pi)$. For all $u, v \in U \cup I$, let

$$p(u, v) := \mathbb{P}'_u(\exists t \geq 1 : X'_1, \dots, X'_{t-1} \in I \setminus \{u\} \text{ and } X'_t = v).$$

Extend this to sets of vertices $L, R \subset V'$ by $p(L, R) := \sum_{u \in L, v \in R} p(u, v)$, and similarly, $p(u, L) := p(\{u\}, L)$ and $p(L, u) := p(L, \{u\})$. Note that this extended definition is not a probability in general, as the walk could potentially reach multiple vertices in R from multiple vertices in L in a suitable way.

Lemma 6.13. *Let $n \geq 1$. Let $P = (p_{i,j})_{i,j \in [n]}$ be a symmetric matrix of non-negative entries such that $\sum_{j \in [n]} p(i, j) \leq 1$ for all $i \in [n]$. Let Γ be the set of all sequences $\gamma = (\gamma_0, \gamma_1, \dots, \gamma_k)$ in $[n]$ (for any $k \geq 1$) with $\gamma_0 = 1$ such that the unique element $i \in [k]$ satisfying both $\gamma_i = 1$ and $\{\gamma_0, \gamma_1, \dots, \gamma_i\} = [n]$ is $i = k$. For every such sequence γ , define*

$$p(\gamma) := \prod_{i=1}^k p(\gamma_{i-1}, \gamma_i).$$

For each $\varepsilon > 0$, if every non-empty proper subset I of $[n]$ satisfies

$$\sum_{i \in I} \sum_{j \in [n] \setminus I} p(i, j) \geq \varepsilon, \quad (17)$$

then $\delta := \frac{\varepsilon^2}{16e^2}$ satisfies

$$\sum_{\gamma \in \Gamma} p(\gamma) \geq \delta^n.$$

We would like to apply Lemma 6.13 to the matrix $P := (p(u, v))_{u, v \in U}$. By time-reversing the trajectories, we have $p(u, v) = p(v, u)$ whenever $d_u = d_v$, which is for example the case when $u, v \in U$. So P is symmetric, and clearly the entries of P are non-negative. Furthermore, the row sums are at most 1 since, defining $T := \inf\{t \geq 1 : X'_t \notin I \setminus \{u\}\}$, we have $p(u, v) = \mathbb{P}'_u(X'_T = v)$, and the sum over $v \in U$ is therefore at most $\mathbb{P}'_u(T < \infty) \leq 1$ by disjointedness of the events $\{X'_T = v\}_{v \notin I \setminus \{u\}}$.

We claim that for every non-trivial partition $U = L \sqcup R$,

$$p(L, R) \geq \varepsilon_2 := \varepsilon_1^2/64. \quad (18)$$

Indeed, for each $x \in U \cup I$, consider the function

$$F(x) := \mathbb{P}'_x(\exists t \geq 0 : X'_0, \dots, X'_{t-1} \notin L \text{ and } X'_t \in R).$$

Note that this is the *unit voltage* as in Remark 5.6, where $a = R$ with $v(R) = 1$, and $Z = L$.

Given $u \in L$, there exists $x \in A$ such that $\{u, x\} \in E'$: if $u \in m(\Pi)$, take x to be the endpoint in A of the cutset edge which has midpoint u ; otherwise take $x = o$. Then if $F(x) \geq 1/2$, then we are done because, by the Markov property,

$$p(u, R) \geq \mathbb{P}'_u(X'_1 = x) \cdot F(x) \geq 1/2 \cdot 1/2 \geq \varepsilon_2. \quad (19)$$

Hence, we may assume that there exists $x \in A$ with $F(x) < 1/2$. Recall that $p(L, R) = p(R, L)$ so by a similar argument (swapping the roles of L and R and thus replacing F by $1 - F$), we may assume that there exists $y \in A$ with $F(y) > 1/2$. Since A is connected in G , x and y are connected by a finite path and so we can find consecutive vertices (endpoints of an edge) $\{x', y'\} \in E$ satisfying $F(x') \leq 1/2 \leq F(y')$. Let $z := m(\{x', y'\})$, which has degree 2. Note that, since $z \in I$ and $y' \in A$ (thus $y' \notin U$),

$$F(z) \geq \mathbb{P}'_z(X'_1 = y') \cdot F(y') \geq 1/2 \cdot 1/2, \quad (20)$$

and by a union bound and the strong Markov property,

$$F(z) \leq \sum_{n=0}^{\infty} \mathbb{P}'_z(\tau > 0)^n p(z, R) = \frac{p(z, R)}{\mathbb{P}'_z(\tau = 0)} \leq \frac{p(z, R)}{\varepsilon_1/2}, \quad (21)$$

where the last inequality follows from (14).

So by rearranging, $p(z, R) \geq \varepsilon_1/8$. By a similar argument (as above), we deduce that $p(L, z) \geq \varepsilon_1/8$. Note that for all $u \in L$ and $v \in R$, $p(u, v) \geq p(u, z)p(z, v)$ and thus $p(L, R) \geq p(L, z)p(z, R)$ by summing over u and v , which implies (18).

Remark 6.14. It is necessary to take care when stating this inequality, as it is not as obvious as it may seem. If $o' \notin L$, so that $u \neq o'$ and thus $u \notin I$, then the strong Markov property gives that $p(u, v) \geq p(u, z)p(z, v)$, since both legs of the decomposition (excluding the start of the first leg and the end of the second) are contained in $I \setminus \{u\}$. If instead we did have that $o' \in L$, then $u = o'$ is possible. If $z = o'$, the inequality becomes trivial. Otherwise, the second leg of the above decomposition may revisit u before reaching v since its intermediate vertices are only restricted to $I \setminus \{z\}$ which contains $u = o'$. In this case, we can simply note that $p(u, v) = p(v, u) \geq p(v, z)p(z, u) = p(u, z)p(z, v)$ by symmetry, since $d_z = 2 = d_u = d_v$, and by the above argument, since $o' \notin R$ so $v \neq o'$.

We can now apply Lemma 6.13. Indeed, the matrix P is as described above; $n = |U|$; the sequences in Γ are the paths of the walk with vertex '1' being o' ; and (17) is precisely (18) with lower bound $\varepsilon_1^2/64$.

Therefore, the event $\mathcal{E}$ that the random walk visits every vertex in U then returns to o' before exiting $U \cup I$ satisfies

$$\mathbb{P}'_{o'}(\mathcal{E}) \geq \varepsilon_3^{|U|} \geq \varepsilon_3^{n+1} \quad (22)$$

for some constant $\varepsilon_3 > 0$ depending only on ε_2 . Now, we can apply Lemma 6.11: on the event $\mathcal{E} \cap \{\tau = 0\}$, the set $C \cap V$ satisfies $B \subset C \cap V \subset A$: the first inclusion holds since the walk visits every vertex of $m(\Pi)$ on $\mathcal{E}$, forcing it to pass through each inner vertex of Π ; the second holds since $V \cap (U \cup I) = A$ and the event $\{\tau = 0\}$ (restarting the walk from its last visit to o' after $\mathcal{E}$ occurs) forces $C \subset (U \cup I)$. Lemma 6.11 then gives $\partial_\infty(C \cap V) = \Pi$ in G . Moreover, since $m(\Pi) \subset C$ and the outer endpoints of Π -edges lie outside $U \cup I$ and hence outside C , the exposed boundary of C in G' consists precisely of the edges between $m(\Pi)$ and these outer endpoints, giving $\partial = m(\Pi)$.

So, by the strong Markov property, (14), and (22),

$$\mathbb{P}'_{o'}(\partial = m(\Pi)) \geq \mathbb{P}'_{o'}(\mathcal{E}) \cdot \mathbb{P}'_{o'}(\tau = 0) \geq \varepsilon_3^{n+1} \cdot \varepsilon_1/2. \quad (23)$$

By expanding the definitions of $\varepsilon_1, \varepsilon_2, \varepsilon_3$ we deduce that (16) holds with $K := \frac{2^{32}(4+\varepsilon)^9}{\varepsilon^9}$. $\square$

We omit the proof of Lemma 6.13 due to space constraints - the full proof can be found in [EST25].

6.4 Dependent Parameters of the Minimal Cutset Bounds

A key aim of this project is to keep track of the parameters on which the exponential upper bounds on the number of minimal cutsets (and hence the resulting upper bounds on p_c^*) depend. We have shown these dependencies within the proofs of each theorem, but we make them slightly more explicit here.

Recall from the proof of Theorem 6.3 that if $\theta > 0$ and $p < 1$ are such that $\mathbb{P}_p(u \leftrightarrow \infty) \geq \theta$, then $|Q_n(o)| \leq K^n$ for any $o \in V$, where $K = (p\theta/2)^{-3/\theta}(1-p)^{-1}$. Clearly, this depends only on p and θ , and was a direct result of Proposition 6.12: the first factor arises from the assumption of a uniform lower bound θ on the probability of an origin vertex connecting to the boundary vertices of its connected component, and the second factor is due to the requirement that all cutset edges are closed. The fact that θ is uniform over all vertices and their connected components is what allows K to depend only on these parameters.

During the proof of Theorem 6.4, it was made explicit that K depends only on ε . Here we simply verify the value of K for which $|Q_n(o)| \leq K^n$:

$$\begin{aligned} \mathbb{P}'_{o'}(\partial = m(\Pi)) &\stackrel{(23)}{\geq} \varepsilon_3^{n+1} \cdot \varepsilon_1/2 \stackrel{(6.13)}{=} \left(\frac{\varepsilon_1^2}{26}\right)^{2n+2} \cdot \frac{1}{16^{n+1}e^{2n+2}} \cdot \frac{\varepsilon_1}{2} = \frac{\varepsilon_1^{4n+5}}{2^{16n+17}e^{2n+2}} \stackrel{(14)}{=} \frac{\left(\frac{2\varepsilon}{4+\varepsilon}\right)^{4n+5}}{2^{16n+17}e^{2n+2}} \\ &= \frac{\left(\frac{\varepsilon}{4+\varepsilon}\right)^{4n+5}}{2^{12n+12}e^{2n+2}} \geq \frac{\left(\frac{\varepsilon}{4+\varepsilon}\right)^{4n+5}}{2^{16n+16}} \geq \frac{\left(\frac{\varepsilon}{4+\varepsilon}\right)^{4n}}{2^{16n}} \cdot \left(\frac{\left(\frac{\varepsilon}{4+\varepsilon}\right)^5}{2^{16}}\right)^n = \left(\frac{\varepsilon^9}{2^{32}(4+\varepsilon)^9}\right)^n, \end{aligned}$$

and so to deduce (16), we can take $K := \frac{2^{32}(4+\varepsilon)^9}{\varepsilon^9}$. Note that this constant differs slightly from [EST25], but the value itself is not important.

6.5 Critical Threshold Gap at 1

As mentioned in Remark 6.9, there exists a universal constant $\varepsilon > 0$ such that every infinite, connected, locally finite, transitive graph G satisfies either $p_c = 1$ or $p_c \leq 1 - \varepsilon$. We split the proof of this into two cases: when G is recurrent and when G is transient. In the former case, we appeal to Theorem 1.7 in [HT24] and, in particular, one passage of the proof.

A result of Varopoulos says that arbitrary, connected, vertex-transitive graphs with *super-quadratic volume growth* are transient [TT24]. That is, by contrapositive, if G is recurrent, then $|B(o, n)| = O(n^2)$ for all $o \in V$, where $|B(o, n)|$ denotes the graph-distance ball of radius n around a fixed vertex o . This notion of super-quadratic volume growth is analogous to that of *superlinear* volume growth, defined in [HT24] as the property that $\limsup_{n \rightarrow \infty} \frac{1}{n} |B(o, n)| = \infty$.

The proof of Theorem 1.7 in [HT24] then splits the cases depending on whether $m := \sup\{n \in \mathbb{N} : |B(o, n)| \geq n^5\} < \infty$ or not. But clearly this is true if G is recurrent by the above discussion. The proof continues by utilising multiple of their own group-theoretic theorems to conclude that there exists a constant M , depending only on the 5 in the above definition of m , such that if G is of superlinear volume growth, then $p_c(G) \leq 1 - \varepsilon(M)$. If G is not of superlinear volume growth, then $p_c(G) = 1$ by the same Theorem 1.7, regardless of recurrence or transience. We conclude the case that G is recurrent by noting that, whether or not G is of superlinear volume growth, G satisfies either $p_c = 1$ or $p_c \leq 1 - \varepsilon$ where ε is a universal constant.

We now proceed in the case where G is transient. Corollary 1.3 in [TT24] immediately gives that, in fact, G is uniformly transient. More specifically, it says that there exists a universal constant $c > 0$ which bounds from below the escape probability of the random walk. We may simply follow the proof of Theorem 6.4 with $\varepsilon = c$, observing that $\kappa(G) \leq K := K(c)$ where K is a finite non-zero constant depending only on c and is thus also universal. Now, in the Peierls direction of Theorem 6.3, the sum in equation (12) converges and is less than 1 when $K(1 - p) < 1/2$, or equivalently, when $p > 1 - 1/2K$. In this case, $\theta^*(p) > 0$ so $p_c^* \leq 1 - 1/2K$ and we are done, since K is universal and Remark 6.6 says that the uniform critical parameter coincides with the non-uniform one since G is transitive.

Out of interest, we observe that the Varopoulos result used in the recurrent case and the approach taken in the transient case are actually related by isoperimetric dimension. A classical result of Coulhon and Saloff-Coste [CS93] says that super-quadratic volume growth implies $\text{Dim}(G) > 2$ [HT24]. We saw earlier in Theorem 5.16 that $\text{Dim}(G) > 2$ implies uniform transience, so in fact, we know a stronger result than Varopoulos: graphs with super-quadratic volume growth are uniformly transient. However, if one follows the proof of Theorem 5.16, they would see that the uniform constant derived as an upper bound on the effective resistance depends on β and d from Remark 5.14; in particular, β and d are graph dependent and can be arbitrarily small and arbitrarily close to 2 from above, respectively. Hence, the uniform constant is not universal over all transient graphs, which is what Remark 6.9 requires. Therefore, we require the additional machinery of [TT24] Corollary 1.3 to know that a universal constant exists.

7 Extension of [EST25] to Site Percolation

It is natural to desire the results we have just seen in the case of site percolation rather than bond percolation. Of particular interest is to obtain a uniform non-trivial upper bound on p_c for site percolation on vertex-transitive graphs, extending the result which is known for *Cayley graphs* (graphs whose vertex set is generated by some set S where an edge uv exists if $u = vs$ for some $s \in S$) [PS23].

In the case of the Cayley graph, the result is proved using an object called the *Gaussian free field*, a natural generalisation of Brownian motion with time replaced by a multidimensional continuous parameter [WP21]. We will not worry too much about the details, as instead, we approach the question from the same perspective as Section 6, and hope to deduce a remark analogous to Remark 6.9.

One may initially think back to Section 3.5 and, in particular, Theorem 3.6, and ask whether the bound on p_c^{site} could help us. Although it is a notable intermediate step, we can only deduce our desired result with a constant ε dependent on the maximum vertex degree of the graph. Hence, it is justifiable to pursue an adaptation of [EST25] to site percolation in order to avoid this degree dependence. We begin by modifying the preliminary definitions.

A natural preceding question is whether the entire site percolation analogue of Theorem 6.3 - not just the Peierls direction (the only part of Theorem 6.3 necessary in Remark 6.9) - is true. We investigate this first, beginning with analogous definitions.

Definition 7.1. Let $G = (V, E)$ be an infinite, connected, locally finite graph. We call a set of vertices $W \subset V$ a *vertex-cutset* from a vertex v to ∞ if v belongs to a finite connected component of $(V \setminus W, E')$ where E' is the set of edges $uv \in E$ such that $u, v \in V \setminus W$. A cutset is called *minimal* if no proper subset of it is a cutset.

Let $Q_n^{\text{site}}(v)$, q_n^{site} , $\kappa^{\text{site}}(G)$, and $p_c^{\text{site}}(G)$ be defined analogously to the bond percolation definitions given in Section 6.1.

7.1 Theorem 1 Extension

Conjecture 7.2. For every infinite, connected, locally finite graph G we have

$$p_c^{\text{site}}(G) < 1 \iff \kappa^{\text{site}}(G) < \infty.$$

Equivalently,

$$(\exists p < 1 \ \exists \theta > 0 \ \forall u \in V \ \mathbb{P}_p^{\text{site}}(u \leftrightarrow \infty) \geq \theta) \iff (\exists K < \infty \ \forall u \in V \ \forall n \geq 1 \ |Q_n^{\text{site}}(u)| \leq K^n).$$

Proof of ($\Leftarrow$). Once again, this implication follows from the Peierls argument. Let $u \in V$ which we may assume is open, because conditioning on this event gives that $\mathbb{P}_p^{\text{site}}(u \leftrightarrow \infty) = p \cdot \mathbb{P}_p^{\text{site}}(u \leftrightarrow \infty | u \text{ open})$, so we can consider θ as having absorbed the constant p . If its cluster C is finite, then the set of vertices $\partial_\infty^V C := \{v \in V \setminus C : v \sim C \text{ and } v \leftrightarrow \infty \text{ in the graph induced by } V \setminus C\}$ is a finite minimal vertex-cutset from u to ∞ .

Indeed, any path from u to ∞ in V must visit a vertex in $\partial_\infty^V C$ (consider the first vertex visited by this path outside of C , after its final visit to C). Therefore, $\partial_\infty^V C$ is a cutset from u to ∞ . To prove that it is minimal, consider a vertex $v \in \partial_\infty^V C$. By definition of $\partial_\infty^V C$, since C is connected, there exists a path from u to a neighbour w of v in C and there also exists a path from v to ∞ in $V \setminus C$. Then in the graph induced by $V \setminus (\partial_\infty^V C \setminus \{v\})$, there exists a path from u to ∞ being the concatenation of these two paths along with the edge between v and w , which exists since both v and w are included in this set of vertices. Hence, $\partial_\infty^V C \setminus \{v\}$ is not a cutset from u to ∞ , and we have shown Lemma 6.10 analogously for vertex-cutsets.

Furthermore, all vertices in $\partial_\infty^V C$ are closed, so by the union bound, for every $p \in [0, 1]$ we have

$$\mathbb{P}_p^{\text{site}}(|C| < \infty) \leq \sum_{n \geq 1} q_n^{\text{site}}(1 - p)^n. \quad (24)$$

The same reasoning as in Section 6.2 applies and so we are done with the Peierls direction. $\square$

7.1.1 Proof of ($\Rightarrow$)

We would like to be able to simply replicate the original proof, but with adaptations for the site percolation setting. As before, fix $\theta, p \in (0, 1)$ such that $\mathbb{P}_p^{\text{site}}(u \leftrightarrow \infty) \geq \theta$ for every $u \in V$. Fix $o \in V$ and $n \geq 1$. Writing C for the cluster of o , we would like to show that for every minimal vertex-cutset Π from o to ∞ with $|\Pi| = n$,

$$\mathbb{P}_p^{\text{site}}(\partial_\infty^V C = \Pi) \geq 1/K^n, \quad (25)$$

where $K = K(p, \theta) \in (0, \infty)$ is a finite constant depending on p and θ only. This concludes the proof since

$$1 \geq \sum_{\Pi \in Q_n^{\text{site}}(o)} \mathbb{P}_p^{\text{site}}(\partial_\infty^V C = \Pi) \geq |Q_n^{\text{site}}(o)|/K^n.$$

Let A be the connected component of o in the graph induced by $V \setminus \Pi$, and let B be the set of neighbours of Π inside A . Any infinite open path from a vertex $u \in A$ must intersect B before exiting A , so the hypothesis $\mathbb{P}_p^{\text{site}}(u \leftrightarrow \infty) \geq \theta$ implies that for all $u \in A$, $\mathbb{P}_p^{\text{site}}(u \xleftrightarrow{A} B) \geq \theta$.

Similarly to the original proof in the bond percolation case, we would like to choose events $\mathcal{E}$ and $\mathcal{F}$ such that on the event $\mathcal{E} \cap \mathcal{F}$, the exposed boundary $\partial_\infty^V C$ of the cluster C of o is forced to equal the minimal cutset Π . In fact, it is enough to let $\mathcal{E}$ be the event that every vertex in Π has a neighbour in A that is connected to o by an open path in A , and to let $\mathcal{F}$ be the event that every vertex of Π is closed.

Indeed, for each $w \in \Pi$, $w \notin C$ and w has a neighbour in C by $\mathcal{E}$. Furthermore, w is connected to ∞ in $V \setminus C$: by minimality of Π there exists a path from $o \rightarrow \infty$ in $V \setminus (\Pi \setminus \{w\})$ which must pass through w , so there exists a path from w to ∞ in $V \setminus A$ which is included in $V \setminus C$. Thus, $w \in \partial_\infty^V C$.

Since C is finite by $\mathcal{F}$, the Lemma 6.10 analogue proved above means that $\partial_\infty^V C$ is a finite minimal vertex-cutset. But then, by the inclusion above and since Π itself is a vertex-cutset, the inclusion must be an equality. That is, $\partial_\infty^V C = \Pi$. Then, all that remains is to lower-bound $\mathbb{P}_p^{\text{site}}(\mathcal{E} \cap \mathcal{F})$.

We would like to have a *useful* analogue of Proposition 6.12. By ‘useful’, we mean that the lower bound, when applied to our use-case, depends not on the size of the neighbourhood of Π in A , but rather just the size of Π itself. For example:

Claim 7.3 (Proposition 6.12 Analogue). *Let $G = (V, E)$ be a finite, connected graph. Let $B \subset V$, let $\theta, p \in (0, 1]$, and suppose that $\mathbb{P}_p^{\text{site}}(u \leftrightarrow B) \geq \theta$ for every $u \in V$. Then for any finite family of disjoint, non-empty sets $\{B_i\}_{i=1}^m$ that partition B , and for every $o \in V$,*

$$\mathbb{P}_p^{\text{site}}\left(\bigcap_{i=1}^m \{o \leftrightarrow B_i\}\right) \geq c^m, \quad c := f(p, \theta) \quad \text{for some positive function } f.$$

We could then apply Claim 7.3 to the finite subgraph of G induced by A so that $\mathbb{P}_p^{\text{site}}(\mathcal{E}) \geq c^n$, by letting the sets B_i be the neighbourhoods in A of each $v \in \Pi$ (and removing elements from sets where they are repeated to enforce disjointedness), and noting that each vertex in Π has exactly one.

Finally, $\mathcal{E}$ and $\mathcal{F}$ are independent due to the disjointedness of the sets they affect, so

$$\mathbb{P}_p^{\text{site}}(\partial_\infty^V C = \Pi) \geq \mathbb{P}_p^{\text{site}}(\mathcal{E} \cap \mathcal{F}) = \mathbb{P}_p^{\text{site}}(\mathcal{E})\mathbb{P}_p^{\text{site}}(\mathcal{F}) \geq c^n(1-p)^n.$$

which would establish (25) with $K = \frac{1}{c(1-p)}$.

7.1.2 Assuming Maximum Degree

It turns out that Claim 7.3 is more difficult to prove than its bond percolation analogue, Proposition 6.12. We make the intermediate remark that by assuming there exists a constant Δ bounding from above the maximum degree of G , Conjecture 7.2 follows immediately.

Indeed, let each B_i in Claim 7.3 be a distinct singleton in B , so there are $m = |B|$ of them. In this case, the claim is reduced precisely to the direct site percolation analogue of Proposition 6.12, which goes through identically (in fact, the proposition as stated in [EST25] needs only a positively associated percolation measure, which the site percolation measure is). Then, $|B|$ would be bounded above by $n\Delta$ since each cutset vertex would have at most Δ neighbours in A , and so the conjecture would be proved with c depending Δ .

Of course, we seek a proof of the proposition for graphs with finite but unbounded degree, so our investigation continues below.

7.1.3 Difficulties in Producing an Analogous Proof

It turns out that if m is a constant that is not dependent on $|B|$ but only on p and θ , Claim 7.3 is actually false, and this can be seen by the following counterexample.

Counterexample 1: Fix a constant $1 < m < |B|$, fix $p \in (0, 1)$, and let $L \geq 1$. Consider the graph G consisting only of a path $A = (v_1, v_2, \dots, v_{mL})$. Let $B = A$ and let $B_i = (v_{(i-1)L+1}, v_{(i-1)L+2}, \dots, v_{iL})$ for all $i \in \{1, 2, \dots, m\}$. Let $o = v_1$ and note that for all $u \in V = A$, $\mathbb{P}_p^{\text{site}}(u \leftrightarrow B) = p$ since as long as the vertex u is open, it is contained in B . Note that since G is a path, the only route from o to B_m is through all intermediate vertices in the path. Hence, given that o is open,

$$\mathbb{P}_p^{\text{site}}(\{o \leftrightarrow B_1\} \cap \dots \cap \{o \leftrightarrow B_m\}) \leq \mathbb{P}_p^{\text{site}}(\{o \leftrightarrow B_m\}) = p^{(m-1)L},$$

and for any $c = f(p, \theta)$ and any fixed m , there exists $L > 0$ such that $p^{(m-1)L} < c^m$, disproving the claim. Note that this is indeed a valid counterexample because, in this example, the constant θ from the statement is p which is independent of the graph. Furthermore, the constant c should indeed be independent of the graph for this proposition to be useful, as the graph to which it is applied depends on the cutset, and we wish for c to be independent of the cutset.

One may instead attempt to avoid the partition structure of Claim 7.3 by reformulating the statement, opting for a more direct bound on $\mathbb{P}_p^{\text{site}}(\mathcal{E})$. Recall that for $\mathcal{E}$ to happen, it is necessary that for every vertex v in Π , there exists an open path from o to a neighbour of v in A . Hence, the most minimal yet sufficient statement possible is the following, where this would be applied to the finite graph induced by $A \cup \Pi$, taking $B = \Pi$.

Claim 7.4. *Let $G = (V, E)$ be a finite, connected graph. Let $B \subset V$, let $\theta, p \in (0, 1]$, and suppose that $\mathbb{P}_p^{\text{site}}(u \overset{V \setminus B}{\leftrightarrow} B) \geq \theta$ for every $u \in V$. Then for every $o \in V$,*

$$\mathbb{P}_p^{\text{site}}\left(\bigcap_{b \in B} \{o \overset{V \setminus B}{\leftrightarrow} b\}\right) \geq c^{|B|}, \quad c := f(p, \theta) \quad \text{for some function } f.$$

The notation $u \overset{V \setminus B}{\leftrightarrow} B$ denotes connection via a path with intermediate vertices in $V \setminus B$. However, this claim is also false, and the previous counterexample shows why.

Counterexample 2: Let A , B_i and o be as in Counterexample 1, and let $G = A \cup \Pi$, where $\Pi = \{\pi_1, \dots, \pi_m\}$ and there is an edge between π_i and every vertex in B_i , for every $1 \leq i \leq m$. Then, similar to above, for any $u \in V \setminus B$, $\mathbb{P}_p^{\text{site}}(u \overset{V \setminus B}{\leftrightarrow} B) = p$ since u is a neighbour of B . But, again, connecting o to B_m through an open path in $V \setminus B$ means all vertices in $B_1, \dots, B_{m-1}$ must be open. Since $|B| = |\Pi| = n$, the same limiting argument above disproves the statement.

7.1.4 Inferences on the Underlying Proof Strategy

The core aim of the proof is to show (25). The idea is to decompose the event $\{\partial_\infty^V C = \Pi\}$ into the event intersection $\{\mathcal{E} \cap \mathcal{F}\}$. We saw above that $\{\mathcal{E} \cap \mathcal{F}\} \subset \{\partial_\infty^V C = \Pi\}$, and I claim that the reverse inclusion also holds. Indeed, if $\{\partial_\infty^V C = \Pi\}$ then each of the cutset vertices must be closed as to not be included in the cluster C , giving the event $\mathcal{F}$. Furthermore, since a cutset vertex is an exposed boundary vertex, it must have a neighbour which is contained in C , giving $\mathcal{E}$. Thus, the event decomposition is not just implicative, but equivalent. Then, since $\mathcal{E}$ and $\mathcal{F}$ are independent, we have that $\mathbb{P}_p^{\text{site}}(\partial_\infty^V C = \Pi) = \mathbb{P}_p^{\text{site}}(\mathcal{E})(1 - p)^n$.

I claim that as a result, our counterexamples show that in fact, (25) is not necessarily true, and this proof strategy fails. Indeed, one can consider embedding multiple versions of the finite graph from Counterexample 2 above into $\mathbb{L}^2$. More explicitly, fix $m > 1$ and for the i^{th} element v_i in an enumeration of the

vertices of $\mathbb{L}^2$, embed the graph above (with $L = i$) into $\mathbb{L}^2$ by connecting each of the Π -vertices to v_i . Then, for any vertex u in this graph, $\mathbb{P}_p^{\text{site}}(u \leftrightarrow \infty) \geq p^2 \theta_{\mathbb{L}^2}^*(p)$. This is because u is either in $\mathbb{Z}^2$, is a Π -vertex which is the neighbour of a vertex in $\mathbb{Z}^2$, or is the neighbour of a Π -vertex. In any case, openness of at most two vertices means that u can reach the subgraph $\mathbb{L}^2$, and the inequality follows from the FKG inequality. Therefore, for an appropriate choice of $p < 1$, this new graph can inherit the non-triviality of the uniform percolation probability $\theta_{\mathbb{L}^2}^*(p)$, and thus also has $p_c^{\text{site}} < 1$. Finally, by the construction of this graph, we can find a vertex o and minimal cutset Π (of length m) such that $\mathbb{P}_p^{\text{site}}(\partial_\infty^V C(o) = \Pi) < 1/K^m$ for any constant $K(p) > 0$.

7.1.5 Line Graphs

One class of (possibly unbounded-degree) graphs for which Conjecture 7.2 does hold is the class of *line graphs*. The line graph $L(G)$ of a graph $G = (V, E)$ has vertex set E , that is, a vertex representing each edge of G , and edges between two vertices if their corresponding edges in G share an endpoint. Often, it is called the *intersection graph* of the edge set of G , because vertices represent the edges of G and the edges represent intersections of edges in G .

This ‘intersection’ perspective helps to visualise the local structure of $L(G)$; if there are k edges incident to a vertex v in V , they pairwise share the endpoint v and so the vertices in $L(G)$ representing these k edges form a k -clique in $L(G)$.

Due to its construction, site percolation with parameter p on $L(G)$ is identical to bond percolation with parameter p on G . Indeed, vertex paths in $L(G)$ correspond bijectively to edge paths in G , so removing a vertex (and all edges incident to it) in $L(G)$ amounts to removing the corresponding edge in G . Therefore, Theorem 6.3 immediately gives Conjecture 7.2 for this type of graph.

The local structure of line graphs reinforces our understanding of the challenges in proving Conjecture 7.2 using the original proof method, and prevents the counterexample scenario we saw earlier. Each (cutset) vertex x in $L(G)$ belongs to at most two cliques, and the neighbourhood of x is contained in the union of these cliques. Therefore, prescribing exposed vertex boundaries is ‘cheaper’ in a line graph: once an open path from a vertex o reaches any neighbour of the cutset vertex x , it can reach any other neighbour of x in at most two additional steps, since those neighbours lie inside a union of at most two cliques.

In the counterexamples above, the graphs were arranged so that reaching the neighbourhood of each cutset vertex required traversing a long sequence of distinct vertices in a path; hence, realising the boundary event required the openness of many intermediate vertices, the number depending on the size of the neighbourhoods. In a line graph, there is no such requirement due to the clique structure of the neighbourhood. Consequently, the number of necessarily enforced open vertices (to prescribe the boundary) grows linearly with the size of the cutset rather than the size of its neighbourhood, indicating that the original proof strategy would still work in this case.

We conclude this section by remarking that, in fact, the class of line graphs is completely characterised by a result of Beineke [Bei70]: a graph is a line graph if and only if it contains none of nine specific graphs as an induced subgraph. In particular, membership of this class is decidable in polynomial time. The presence of any such forbidden subgraph introduces local structure incompatible with the clique cover property, which is precisely what prevents our previous counterexample from applying to line graphs.

7.2 Alternative Strategies and Theorem 2 Analogue

It is evident from our previous discussion that prescribing vertex boundaries proves difficult. A similar approach is used in the proof of Theorem 6.4, so it is natural to question whether the random walk argument makes it any easier to replicate in the site percolation analogue.

Conjecture 7.5. *Let G be an infinite, connected, locally finite graph. If G is uniformly transient, then $\kappa^{\text{site}}(G) < \infty$.*

Using the exact same notation as in Section 6.3, let ∂ now denote the exposed vertex boundary of $C \cap V$ in G . Fix $o \in V$, and pick an edge-midpoint neighbour o' of o in G' . Fix a finite minimal vertex cutset Π from o to ∞ in G , and set $n := |\Pi|$.

The aim is to show that for some finite constant $K \in (0, \infty)$ depending only on ε ,

$$\mathbb{P}'_{o'}(\partial = \Pi) \geq 1/K^n. \quad (26)$$

This implies that $\kappa^{\text{site}}(G) < \infty$ for the exact same reason as for (16). However, we claim that (26) is in fact not true, and that a similar counterexample as in Section 7.1.4 applies.

Indeed, let $m > 1$ and $L \geq 1$ be constants. Consider the graph H_L consisting of the union of a path $A_L = (v_1, v_2, \dots, v_{mL})$ and the enumerated set Π_L of m vertices such that, for all $i \in \{1, 2, \dots, m\}$, the i^{th} element of Π_L is connected only to every vertex in the set $B_L^{(i)} = (v_{(i-1)L+1}, v_{(i-1)L+2}, \dots, v_{iL})$.

Now consider the graph G constructed by enumerating the vertices of $\mathbb{L}^3$ and, for all $i \in \mathbb{N}$, embedding the graph H_i into $\mathbb{L}^3$ by simply connecting each of the vertices of $\Pi_i \subset V(H_i)$ to the i^{th} vertex in the enumeration of the vertices of $\mathbb{L}^3$. Note that this construction is almost identical to the graph constructed in Section 7.1.4, except that the underlying graph is $\mathbb{L}^3$ rather than $\mathbb{L}^2$.

G is infinite, connected, and locally finite. Moreover, it is uniformly transient - a property inherited from $\mathbb{L}^3$. Indeed, suppose a vertex $v \in V(G)$ is part of the $\mathbb{L}^3$ subgraph. Then by Rayleigh's Monotonicity Principle [LP17], since the conductances in G are at least the conductances in $\mathbb{L}^3$ (thinking of this as G with zero conductance on edges that are outside of the $\mathbb{L}^3$ subgraph), $\mathcal{C}_G(v \leftrightarrow \infty) \geq \mathcal{C}_{\mathbb{L}^3}(v \leftrightarrow \infty) > C$ for some constant $C > 0$, since $\mathbb{L}^3$ is uniformly transient. Now, every vertex u in G that is not part of the $\mathbb{L}^3$ subgraph is either a cutset vertex, or adjacent to a cutset vertex π which is adjacent to a vertex $a \in V(\mathbb{L}^3)$. In either case, $\mathcal{R}_G(u \leftrightarrow \infty) \leq \mathcal{R}_G(u \leftrightarrow \pi) + \mathcal{R}_G(\pi \leftrightarrow a) + \mathcal{R}_G(a \leftrightarrow \infty) \leq 2 + 1/C$ by the triangle inequality from Remark 5.11, and since the direct edge (of weight 1, assuming u is as far away from the $\mathbb{L}^3$ subgraph as possible) between u and π , and π and a , respectively, guarantees effective resistance for each pair to be at most 1 by Rayleigh's Monotonicity Principle again. Therefore, $\mathcal{C}_G(u \leftrightarrow \infty) \geq (2 + 1/C)^{-1} > 0$ and so, regardless of the vertex picked in G , its effective conductance from ∞ is uniformly bounded away from zero and so G is uniformly transient.

We claim that with $n = m$ and G' constructed as usual from G , (26) fails. That is, if we let o_i be the first vertex in A_i and o'_i be the midpoint between the first and second vertices of A_i , then we claim that there exists large enough i such that $\mathbb{P}'_{o'_i}(\partial = \Pi_i) < 1/K^m$ for any constant K depending only on the uniform transience constant of G' .

Indeed, for $\partial = \Pi_i$ to hold, the walk on G' must reach block $B_m^{(i)}$ before time τ without ever visiting any vertex of Π_i before τ . In particular, it must traverse at least $(m-1)i-1$ intermediate (pre-graph-splitting) path vertices before reaching $B_m^{(i)}$, since the path structure of the gadget forces the walk to visit each in order. Let $T_2 < T_3 < \dots < T_{(m-1)i}$ denote the successive first-visit times to these intermediate path vertices (noting that the walk starts between the first and second vertices of the first block, and so the simplest path is through the second one first), and note that they are all finite on the event $\partial = \Pi_i$. Each such vertex w_k has degree 3 in G' , with neighbours consisting of its two (degree-2) path neighbours and the midpoint $m(\{w_k, \pi_j\})$ of the unique cutset edge incident to it. Given $T_k < \infty$, by the strong Markov property, the walk then steps toward π_j with probability $\frac{1}{3}$, and from there reaches π_j with probability $\frac{1}{2}$, so the probability of reaching π_j within the next two steps from w_k is exactly $\frac{1}{6}$.

If the walk ever reaches a cutset vertex π_j before time τ , then $\pi_j \in C \cap V$ and $\partial \neq \Pi^{(i)}$. Therefore $\{\partial = \Pi_i\} \subseteq \bigcap_{k=2}^{(m-1)i} (E_k^c \cap \{T_k < \infty\})$, where E_k denotes the event that the walk reaches the cutset vertex

(2-step) neighbour of w_k within two steps of time T_k . Hence, we have that

$$\begin{aligned}
\mathbb{P}'_{o'_i}(\partial = \Pi_i) &\leq \mathbb{P}'_{o'_i} \left(\bigcap_{k=2}^{(m-1)i} (E_k^c \cap \{T_k < \infty\}) \right) \\
&= \mathbb{P}'_{o'_i}(E_2^c \cap \{T_2 < \infty\}) \prod_{k=3}^{(m-1)i} \mathbb{P}'_{o'_i} \left((E_k^c \cap \{T_k < \infty\}) \left| \bigcap_{j=2}^{k-1} (E_j^c \cap \{T_j < \infty\}) \right. \right) \\
&\leq \mathbb{P}'_{o'_i}(E_2^c) \prod_{k=3}^{(m-1)i} \mathbb{P}'_{o'_i} \left(E_k^c \left| \{T_k < \infty\} \cap \bigcap_{j=2}^{k-1} (E_j^c \cap \{T_j < \infty\}) \right. \right) \\
&= \mathbb{P}'_{o'_i}(E_2^c) \prod_{k=3}^{(m-1)i} \mathbb{P}'_{w_k}(E_k^c) = \left(\frac{5}{6} \right)^{(m-1)i-1},
\end{aligned}$$

where the first equality uses the probabilistic chain rule; the next inequality uses inclusion of events and the definition of conditional expectation; and the penultimate inequality uses the strong markov property and independence/measurability. Thus, for any constant K depending only on the uniform transience constant of G' , one can find o'_i with i large enough such that $\mathbb{P}'_{o'_i}(\partial = \Pi_i) < 1/K^m$, and so (26) fails.

7.2.1 Assuming Maximum Degree

We remark, similarly to in Section 7.1.2, that Conjecture 7.5 follows easily if we assume that G has maximum degree at most $\Delta > 0$.

Indeed, each minimal vertex cutset Π of size n generates an edge cutset (the edges between A and Π) of size at most $n\Delta$. This edge cutset contains a minimal edge cutset of size at most $n\Delta$. Conversely, a given edge cutset of size k determines at most 2^k possible vertex cutsets. Therefore,

$$|Q_n^{\text{site}}(o)| \leq \sum_{k=1}^{n\Delta} |Q_k(o)| \cdot 2^k \leq \sum_{k=1}^{n\Delta} K^k \cdot 2^k = \frac{2K((2K)^{n\Delta} - 1)}{2K - 1} \leq \frac{2K}{2K - 1} \cdot (2K)^{n\Delta} \leq (2K)^{2n\Delta},$$

so $\kappa^{\text{site}}(G) \leq (2K)^{2\Delta} < \infty$.

We originally desired Conjecture 7.5 in order to reach a site percolation analogue of Remark 6.9 - that there is a ‘gap at 1’ for the critical parameter. The case when G is transient would follow analogously as in Section 6.5, and the case when G is recurrent requires more care, which we will visit in Section 8.

However, we note that there is a much simpler way to deduce Remark 6.9 for the class of graphs with maximum degree at most $\Delta > 0$, instead of first proving Conjecture 7.5 for such graphs. Indeed, let G be an infinite, connected, locally finite, vertex transitive graph with maximum degree at most $\Delta > 0$, and assume that $p_c^{\text{site}}(G) < 1$. Then by Theorem 3.6, $p_c^{\text{bond}}(G) < 1$ so by Remark 6.9, $p_c^{\text{bond}}(G) < 1 - \varepsilon$ for some $\varepsilon > 0$. By the same theorem, we have that $p_c^{\text{site}}(G) < 1 - \varepsilon^\Delta$. Hence, ε^Δ is a universal constant for this class of graphs.

7.3 Concluding Remarks

The fundamental obstruction in extending both theorems to site percolation is that prescribing an exposed vertex boundary requires control over the neighbourhoods of the cutset vertices. In the bond percolation proofs, paths from the origin to inner endpoints of cutset edges may freely pass through other inner endpoints; in the site setting, paths to cutset vertex neighbourhoods must avoid the cutset vertices, coupling the problem to the geometry between neighbourhoods. The total number of neighbourhood

vertices to be controlled is degree-dependent and unbounded relative to the cutset size, which is precisely where both proof strategies in the site percolation setting break down.

Despite these negative results, we emphasise that our counterexamples do not disprove Conjectures 7.2 or 7.5 themselves. By construction, the counterexample graphs have very few additional cutsets of the chosen size, and they satisfy the conjectures since their underlying lattice has bounded degree. A proof of either conjecture must therefore exploit a global relationship between the number of cutsets and the structure of the graph - one that tolerates individual cutsets being ‘expensive’ to prescribe, provided they are sufficiently few.

Possible future directions for a proof of Conjecture 7.5 may involve using the Gaussian Free Field, as is used in [PS23], since we have seen that tools beyond probabilistic cutset counting may be better suited to the site percolation setting. Furthermore, the observation that our counterexamples have very few cutsets of the chosen size hints that a proof of the conjectures in full generality could also proceed by showing that graphs admitting many minimal vertex-cutsets of size n necessarily have local geometry favourable enough to offset the cost of prescribing any individual one. For instance, one idea could be to relate the cutset count to the isoperimetric profile or effective resistance structure of the connected components they induce, since in our counterexamples, both properties are extremal.

8 Deducing a Corollary That Extends Current Results

We conclude this report with an original contribution in the form of a deduction of the partial resolution of conjectures posed by Panagiotis and Severo [PS23], and Benamini and Schramm [BS96]. Panagiotis and Severo conjecture that every infinite, connected, (quasi-)transitive graph G of superlinear volume growth admits a uniform $\varepsilon > 0$ such that $p_c^{\text{site}}(G) < 1 - \varepsilon$. Benamini and Schramm make the weaker conjecture that for such graphs, $p_c^{\text{site}}(G) < 1$. We show that the former statement holds if G is recurrent by similar reasoning to that of Section 6.5, implying the latter statement for the same graph. To do so, we cite the following results, assuming for the sake of brevity that the reader understands some group theory.

[PS23] **Theorem 1.1.** There is a universal constant $\varepsilon > 0$ such that $p_c^{\text{site}}(G) < 1 - \varepsilon$ for any Cayley graph G of superlinear growth.

[BS96] **Theorem 1.** Let G be a locally finite graph and $H < \text{Aut}(G)$. Then $p_c^{\text{site}}(G) \leq p_c^{\text{site}}(G/H)$.

We also state the following slightly modified version of Lemma 2.9 from [HT24] where the proof is identical, so we don’t include it here. In this lemma, we use π to denote the quotient map from V to V/H .

Lemma 8.1. Suppose that $G = (V, E)$ is a locally finite graph and that $H < \text{Aut}(G)$ satisfies $|Hv| \leq k$ for some $k \geq 1$ and every $v \in V$. For each $v \in G$ and $p \in [0, 1]$, the cluster of $\pi(v)$ in Bernoulli- $(1 - (1 - p)^k)$ site percolation on G/H stochastically dominates the image under π of the cluster of v in Bernoulli- p site percolation on G . That is,

$$\mathbb{P}_{1-(1-p)^k}^{G/H}(K_{\pi(v)} \in A) \geq \mathbb{P}_p^G(\pi(K_v) \in A)$$

for every increasing measurable set $A \subseteq \{0, 1\}^{V/H}$.

Now we can state and prove the following partial resolution of the conjectures mentioned above.

Corollary 8.2. There is a universal constant $\varepsilon_0 > 0$ such that $p_c^{\text{site}}(G) < 1 - \varepsilon_0$ for any recurrent, transitive graph G of superlinear growth.

Proof. By similar logic to that of the recurrent case in Section 6.5, and by following the proof of Theorem 1.7 in [HT24], we get that there exist groups $H < \text{Aut}(G)$ and $\Gamma < \text{Aut}(G/H)$ such that: Γ acts transitively on V/H ; there exists a finite symmetric generating set S for Γ ; the growth of $\text{Cay}(\Gamma, S)$ is

superlinear; and G/H is isomorphic to a quotient of $\text{Cay}(\Gamma, S)$ by a subgroup of $\text{Aut}(\text{Cay}(\Gamma, S))$ of order at most M (the same M from Section 6.5).

Then [PS23] Theorem 1.1 applies so that $p_c^{\text{site}}(\text{Cay}(\Gamma, S)) < 1 - \varepsilon$, and Lemma 8.1 implies that $p_c^{\text{site}}(G/H) < 1 - \varepsilon_0$ for some universal constant $\varepsilon_0 > 0$ depending only on M and ε , where we recall that M is independent of the graph and the percolation. Finally, [BS96] Theorem 1 implies that $p_c^{\text{site}}(G) < 1 - \varepsilon_0$, which concludes the proof. $\square$

We suspect that Corollary 8.2 also holds for transient graphs, and endeavor to prove it in the future via a proof of Conjecture 7.5.

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